

SMART GUIDELINES

Digital adaptation kit for child health (0–59 months) in humanitarian emergencies

**Operational requirements for
implementing WHO recommendations
in digital systems**



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Implementation tools

Core data dictionary	smart.who.int/dak-ccc/dictionary
Decision-support logic	smart.who.int/dak-ccc/decision-logic
Indicators and performance metrics	smart.who.int/dak-ccc/indicators
Functional and non-functional requirements	smart.who.int/dak-ccc/system-requirements

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Abbreviations

BMI	body mass index
CHW	community health worker
DAK	digital adaptation kit
DHIS2	District Health Information System 2
DTDS	digital tracking and decision support
EMR	electronic medical record
FHIR	Fast Health Interoperability Resources
HMIS	health management information system
ICD	International Classification of Diseases and Related Health Problems
IMCI	Integrated Management of Childhood Illness
MOH	ministry of health
MUAC	mid-upper arm circumference
SMART	s tandards-based, m achine-readable, a daptive, r equirements-based and t estable
UHC	universal health coverage

Glossary

Business process	A set of related activities or <i>tasks</i> performed together to achieve the objectives of the health programme area, such as registration, counselling and referrals (1, 13).
Data dictionary	A centralized repository of information about the <i>data elements</i> that contains their definition, relationships, origin, use and type of data. For this DAK, the data dictionary is provided as a spreadsheet.
Data element	A unit of data that has specific and precise meaning.
Decision-support logic	A set of decision rules for standard and exceptional cases that is separate from the <i>business process</i> . This would help reduce the complexity of the business process depiction without losing the detail necessary for coding the rules required for system functionality.
Decision-support (for health workers)	Digitized job aids that combine an individual's health information with the health worker's knowledge and clinical protocols to assist health workers in making diagnosis and treatment decisions (7, 8).
Decision-support table	Semi-structured way to depict each discrete decision that will need to be embedded in the system. Depending on the complexity of the clinical guidelines, there will likely be multiple decision-support tables.
Digital health	The systematic application of information and communications technologies, computer science and data to support informed decision-making by individuals, the health workforce and health systems, to strengthen resilience to disease and improve health and wellness (1, 25).
Digital tracking	The use of a digitized record to capture and store clients' health information to enable follow-up of their health status and services received. This may include digital forms of paper-based registers and case management logs within specific target populations, as well as electronic medical records linked to uniquely identified individuals (7, 8).
Functional requirement	Capabilities the system must have to meet the end users' needs and achieve <i>tasks</i> within the <i>business process</i> .
Health information system	A system that integrates data collection, processing, reporting and use of the information necessary for improving health service effectiveness and efficiency through better management at all levels of health services (27).
Health management information system	An information system specifically designed to assist in the management and planning of health programmes, as opposed to delivery of care (27).

Interoperability	The ability of different applications to access, exchange, integrate and use data in a coordinated manner through the use of shared application interfaces and standards, within and across organizational, regional and national boundaries, to provide timely and seamless portability of information and optimize health outcomes.
Non-functional requirement	General attributes and features of the digital system to ensure usability and overcome technical and physical constraints. Examples of non-functional requirements include the ability to work offline, multiple language settings and password protection.
Persona	A generic aggregate description of a person involved in or benefiting from a health programme.
Standard	In software, a standard is a specification used in digital application development that has been established, approved and published by an authoritative organization. These rules allow information to be shared and processed in a uniform, consistent manner independent of a particular application.
Task	A specific action in a <i>business process</i> .
Terminologies	For clinical care, terminologies are structured vocabularies covering health-related concepts, such as diseases, diagnoses, laboratory tests and treatments, to enable the storage, analysis and exchange of data in a consistent and standard way (27).
Workflow	A visual representation of the progression of activities (tasks, events, decision points) in a logical flow illustrating the interactions within the <i>business process</i> (28).

Note: Terms in the definitions also defined in this glossary are shown in *italics*.

1

Part 1. Overview of digital adaptation kits

Background

Digital health – defined broadly as the systematic application of information and communications technologies, computer science and data to support informed decision-making by individuals, the health workforce and health systems, to strengthen resilience to disease and improve health and wellness (1) – is increasingly being applied as an essential enabler for health service delivery and accountability. Ministries of health (MOHs) have recognized the value of digital health as articulated within the World Health Assembly resolution (2) and the *Global strategy on digital health* (3). Likewise, donors have advocated for the rational use of digital tools as part of efforts to expand coverage and quality of services, and promote data use and monitoring efforts (4–6). Despite the investments into and abundance of digital systems, there is often limited transparency in the health data and logic contained in these digital tools, or relationship with evidence-based clinical or public health recommendations, which not only undermines the credibility of such systems, but also impedes opportunities for interoperability and threatens the potential for continuity of care.

Evidence-based recommendations, such as those featured in WHO guidelines, establish standards of care and offer a reference point for informing the content of digital systems that countries adopt. However, guidelines are often only available in a narrative format that requires a resource-intensive process to be elaborated into the specifications needed for digital systems. This translation of guidelines for digital systems often results in subjective interpretation for implementers and software vendors, which can lead to inconsistencies or inability to verify the content within these systems, potentially leading to adverse health outcomes and other unintended effects. Where digital systems exist, the documentation of the underlying data and content may be unavailable or proprietary, requiring governments to start from scratch and expend additional resources each time they intend to deploy such a system. This lack of documentation of the health content can lead to dependence on one vendor and haphazard deployments that are unscalable or difficult to replicate across different settings.

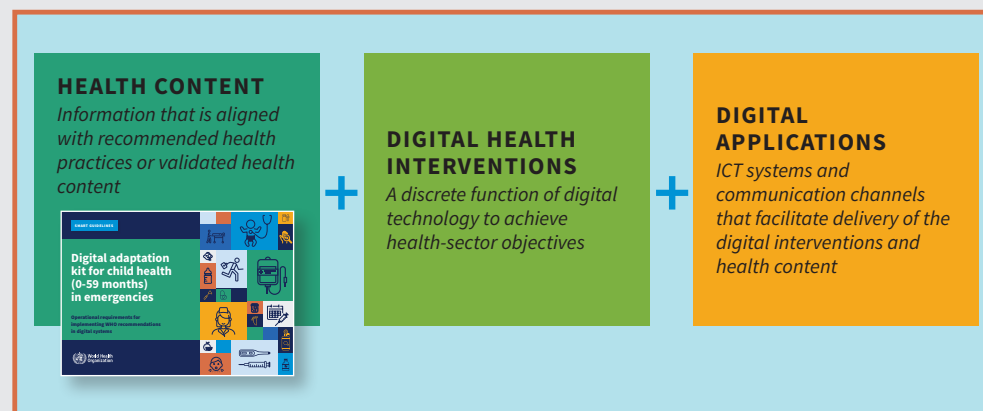
To ensure that countries can effectively benefit from digital health investments, digital adaptation kits (DAKs) are designed to facilitate the accurate reflection of WHO’s clinical, public health and data use guidelines within the digital systems countries are adopting. DAKs are operational, software-neutral, standardized documentation that distil clinical, public health and data use guidance into a format that can be transparently incorporated into digital systems. Although digital implementations consists of multiple factors – including (1) health domain data and content, (2) digital intervention or functionality, and (3) digital application or communication channel for delivering the digital intervention – DAKs focus primarily on ensuring the validity of the health content (Fig. 1) (1, 7). Accordingly, DAKs provide the generic content requirements that should be housed within digital systems, independently of a specific software application and with the intention that countries can customize them to local needs.

For this DAK, the requirements are based on systems that provide the functionalities of digital tracking and decision support (DTDS) (Box 1) and include components such as personas, workflows, core data elements, decision-support algorithms, scheduling logic and reporting indicators. Operational outputs, such as spreadsheets of the data dictionary and the detailed decision-support algorithms, are included as part of the DAK as practical resources that implementers can use as starting points when developing digital systems. Furthermore, data components within the DAK are mapped to standards-based terminology, such as the International Classification of Diseases and Related Health Problems (ICD), to facilitate interoperability.

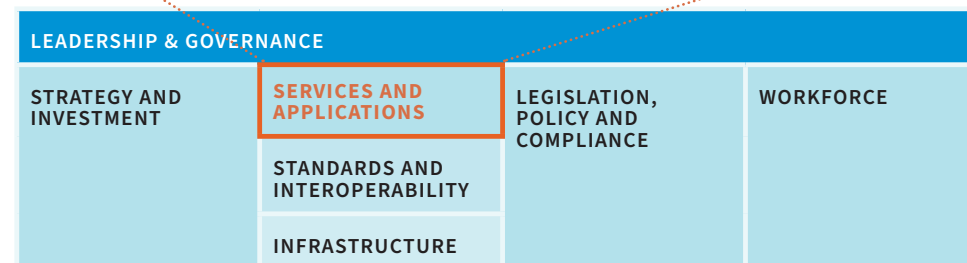
DAKs follow a modular approach in detailing the data and content requirements for a specific health programme area – such as antenatal care and child health among the different health areas for which DAKs have been developed. **This DAK focuses on providing the content requirements for a DTDS system used in primary health care settings by health workers for child health in humanitarian emergencies.**

Fig. 1

Digital adaptation kits and their role within digital health implementations



Foundational Layer: ICT and enabling environment



ICT: information and communications technology.

Box 1

What is DTDS?

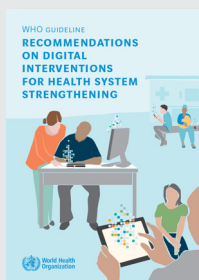
Digital tracking is the use of digitized records to capture and store clients' health information to enable follow-up of their health status and services received (8). This may include digital forms of paper-based registers and case management logs within specific target populations, as well as electronic patient records linked to uniquely identified individuals (7, 8).

Digital tracking makes it possible to register and follow up patient services, and may be done through an electronic medical record or other digital forms of health records. Digital tracking aims to reduce lapses in continuity of care by stimulating timely follow-up contacts, and may incorporate decision-support tools to guide health workers in executing clinical protocols to deliver appropriate care; scheduling upcoming services; and following checklists for appropriate case management at point of care. Some other descriptors include: digital versions of paper-based registers for specific health domains; digitized registers for longitudinal health programmes, including tracking of migrant populations' benefits and health status; and case management logs within specific target populations, including migrant populations (8).

Health worker decision support is defined as digitized job aids that combine an individual's health information with the health worker's knowledge and clinical protocols to assist health workers in making diagnosis and treatment decisions (8). Thus, a person-centred DTDS system is one used by health workers at the point of care; it includes a persistent record of health events and encounters that links to clinical decision-support systems to reinforce good practice. It also links to reporting and management tools to reinforce accountability. A DTDS record includes all the information required for detailing an individual's health status and the health interventions provided to them.

DTDS end users are all cadres of health-care providers operating at all care levels, including those operating outside formal health-care facilities (e.g. community health workers, health volunteers). DTDS systems emphasize the use of "collect once, use for many purposes" (9), in which data collected for service delivery can also be used for accountability (i.e. they can be used to calculate aggregate indicators required for reporting, including monitoring provider, stock and system performance).

WHO has provided the following context-specific recommendation for the use of an integrated system that provides both a digital track of client's health status and decision support (7).



Effective coverage Accountability coverage

Digital tracking of clients' health status and services (digital tracking) combined with decision support

RECOMMENDATION 8: WHO recommends digital tracking of clients' health status and services, combined with decision support under these conditions:

- ▶ in settings where the health system can support the implementation of these intervention components in an integrated manner; and
- ▶ for tasks that are already defined as within the scope of practice for the health worker.

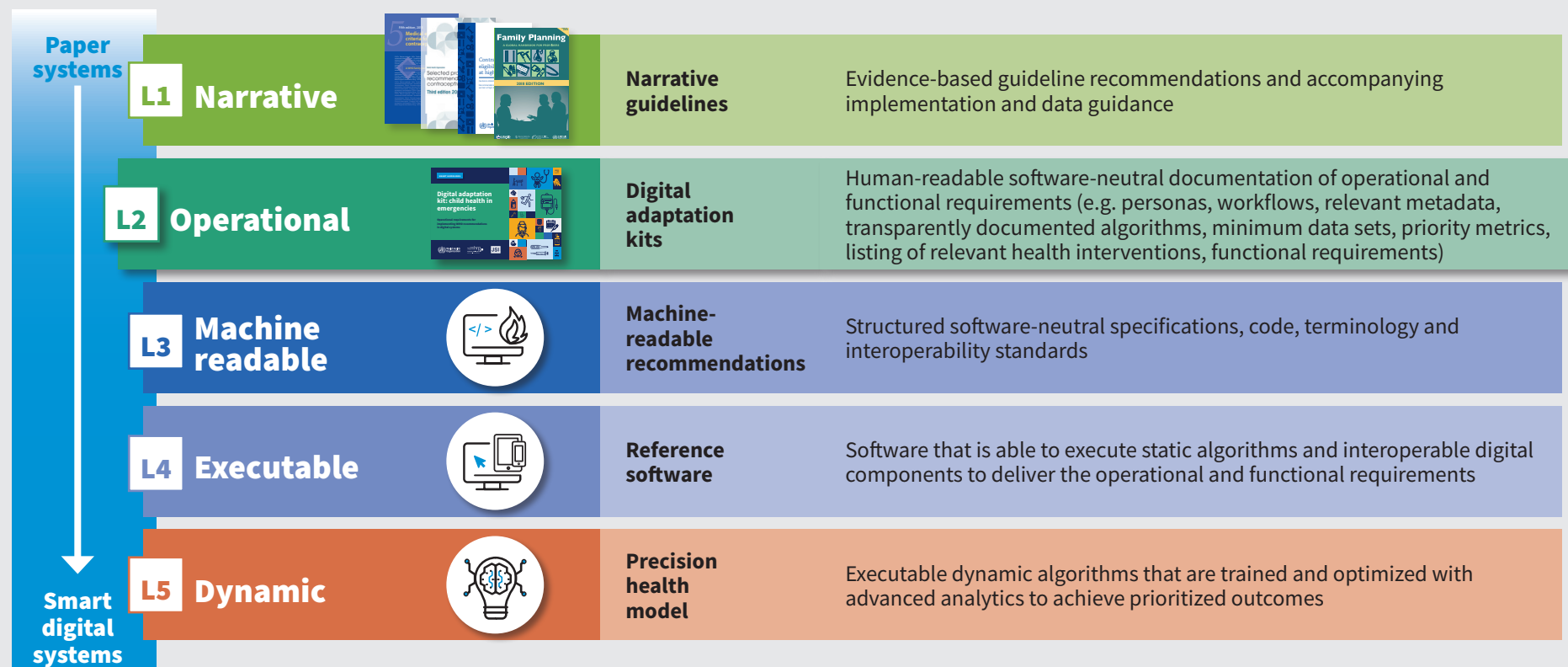
(Recommended only in specific contexts or conditions)

Digital adaptation kits within a strategic vision for SMART guidelines

The operational and standardized documentation reflected within the DAKs represent one of the steps within a broader vision of standards-based, machine-readable, adaptive, requirements-based and testable (SMART) guidelines. SMART guidelines aim to maximize health impact through improved fidelity and uptake of recommendations within standards-based digital systems through a systematic process for transforming guideline development, delivery and application (10, 11). Within this vision, DAKs serve as a prerequisite for developing computable, or machine-readable, guidelines, as well as executable reference software and advanced analytics for precision health. Figure 2 provides an overview of the different layers of the SMART guidelines continuum and where DAKs fit within this strategy (10).

Fig. 2

Progressive layers across SMART Guideline components



Objectives

This DAK focuses on child health in humanitarian emergencies and aims to provide a common language across various audiences – child health in humanitarian emergencies programme managers, software developers and implementers of digital systems – to ensure a common understanding of the appropriate health information content within a defined health programme area, as a mechanism to catalyse the effective use of these digital systems. The key objectives of the DAK are:

- » to ensure adherence to WHO clinical, public health and data use guidelines, and facilitate consistency of the health content that is used to inform the development of a person-centred DTDS system;
- » to enable both health programme leads and digital health teams (including software developers) to have a joint understanding of the health content within the digital system, with a transparent mechanism to review the validity and accuracy of the health content; and
- » to provide a starting point of the core data elements and decision-support logic that should be included within DTDS systems for child health in humanitarian emergencies.

Information detailed in this DAK reflects generic workflow processes, data and decision-support algorithms, as derived from the *Integrated management of childhood illness website* (<https://www.who.int/teams/maternal-newborn-child-adolescent-health-and-ageing/child-health/integrated-management-of-childhood-illness>) and other related WHO documents described below. Note that the outputs of the DAKs are intentionally generic and will need to be contextualized to local policies and requirements.

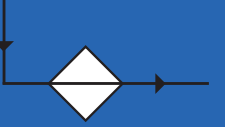
DAKs have also been developed for antenatal care, family planning and HIV, and this approach is being expanded to additional health domains, such as immunizations and postnatal care. All of these DAKs work towards a comprehensive approach for standardized software requirements for primary health care settings.

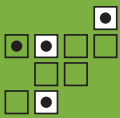
Components of a digital adaptation kit

The DAK consists of eight interlinked components: (1) health interventions and associated recommendations; (2) generic personas; (3) user scenarios; (4) generic business processes and workflows; (5) core data elements; (6) decision-support logic; (7) indicators and reporting requirements; and (8) high-level functional and non-functional requirements. **Table 1** provides an overview of each of these components, which this document elaborates. All information within the DAK represents a generic starting point, which can then be adapted according to the specific context.

Throughout the DAK, there are ID numbers to simplify tracking and referencing of each of the components. Note that the DAK represents an overview across the different components, while the comprehensive and complete outputs of each component (e.g. data dictionary, decision-support tables) are appended as web annexes. The notation guidance is shown in **Box 2**.

Table 1: Components of a digital adaptation kit (DAK)

Component	Description	Purpose	Outputs	Adaptation needed
1. Health interventions and recommendations 	Overview of the health interventions and WHO recommendations included within this DAK. DAKs are meant to be a repackaging and integration of WHO guidelines and guidance documents in a particular health domain. The list of health interventions is drawn from the UHC menu of interventions compiled by WHO (12)	Setting the stage to understand how this DAK would be applied to a DTDS system in the context of specific health programmes and interventions	» List of related health interventions based on WHO's UHC essential interventions » List of related WHO recommendations based on guidelines and guidance documents	» Contextualization to reflect current or planned national policies
2. Generic personas 	Depiction of the end users, supervisors and related stakeholders who would be interacting with the digital system or involved in the care pathway	Contextualization to understand the wants, needs and constraints of the end users	» Description, competencies and essential interventions performed by targeted personas	» Greater specification and details on the end users based on real people (e.g. health workers) in each context » High-level information to describe the provider of the health service (e.g. the general background, roles and responsibilities, motivations, challenges, and environmental factors)
3. User scenarios 	Narratives that describe how the different personas may interact with each other The user scenarios are only illustrative and are intended to give an idea of a typical workflow	Contextualization to understand how the system would be used, and how it would fit into existing workflows	» Example narrative of how the targeted personas may interact with each other during a workflow	» Greater specification and details on the real needs of end users in each context
4. Generic business processes and workflows 	A business process is a set of related activities or tasks performed together to achieve the objectives of the health programme area, such as registration, counselling and referrals (1, 13) Workflows are a visual representation of the progression of activities (tasks, decision points, interactions) that are performed within the business process (1, 13)	Contextualization and system design – To understand how the digital system would fit into existing workflows and how best to design the system for that purpose	» Overview matrix presenting the key processes in child health in humanitarian emergencies » Workflows for identified business processes with annotations	» Customization of the workflows that can include additional forks, alternative pathways or entirely new workflows

Component	Description	Purpose	Outputs	Adaptation needed
5. Core data elements 	Data elements required throughout the different points of the workflow These data elements are mapped to the ICD-11 codes and other established concept mapping standards to ensure that the data dictionary is compatible with other digital systems	System design and interoperability to know which data elements need to be logged and how they map to other standard terminologies (e.g. ICD, SNOMED) for interoperability with other standards-based systems	» List of data elements » Link to data dictionary with detailed data specifications in spreadsheet format here	» Translation of data labels into the local language and additional data elements created depending on the context
6. Decision-support logic 	Decision-support logic and algorithms to support appropriate service delivery in accordance with WHO clinical, public health and data use guidelines	System design and adherence to recommended clinical practice to know what underlying logic needs to be coded into the system	» List of decisions that need to be made throughout the encounter » Link to decision-support tables in spreadsheet format with inputs, outputs and triggers for each decision-support logic here	» Change of specific thresholds or triggers in a logic (IF/THEN) statement, for example, fever reported is recorded as true when the temperature is above a certain threshold or the child is hot to the touch » Additional decision-support logic formulae depending on the context
7. Indicators and performance metrics 	Core set of indicators that need to be aggregated for decision-making, performance metrics, and subnational and national reporting These indicators and metrics are based on data that can feasibly be captured from a routine digital system, rather than survey-based tools	System design and adherence to recommended health monitoring practices to know what calculations and secondary data use are needed for the system, based on the principle of collect once, use many (9)	» Link to indicators table with numerator and denominator of data elements for calculation, along with appropriate disaggregation here	» Changing the calculation formulae of indicators » Adding indicators » Changing the definition of the primary data elements used to calculate the indicator based on the available data
8. High-level functional and non-functional requirements 	List of core functions and capabilities the system must have to meet the end users' needs and achieve tasks within the business process	System design to know what the system should be able to do	» Link to functional and non-functional requirements tables with the intended end user of each requirement, as well as why that user needs that functionality in the system here	» Adding or reducing functions and system capabilities based on budget and end user needs and preferences

DTDS: digital tracking and decision support; ICD-11: International Classification of Diseases and Related Health Problems, 11th revision; SNOMED: Systematized Nomenclature of Medicine; UHC: universal health coverage.

Box 2

Notation guidance

Component 1: Health interventions and recommendations

No notations used.

Component 2: Generic personas

No notations used.

Component 3: User scenarios

No notations used.

Component 4: Business processes and workflows

- » Each workflow should have a process name and a corresponding letter. Each workflow should also have a **process ID** that should be structured as **abbreviated health domain** (e.g. CHE) and **corresponding letter for the process** (e.g. A).
- » Each activity in the workflow should be numbered with an **activity ID** that should be structured as **process ID** (from above) and **activity number** (e.g. CHE.B7).

Component 5: Core data elements (data dictionary)

- » Each data element should have a running number and a **data element (DE) ID** that should be structured as **abbreviated health domain** (e.g. CHE), **activity ID** (e.g. B7) and sequential number of the data element (e.g. CHE.B7.DE.1, CHE.B7.DE.2).

Component 6: Decision-support logic

- » Each decision-support logic table should have a running number and a **decision-support table (DT) ID** that should be structured as **abbreviated health domain** (e.g. CHE) and sequential number of the decision-support table (e.g. CHE.DT.01, CHE.DT.02).

Component 7: Indicators and performance metrics

- » Each indicator should have an **indicator ID** that should be structured as **abbreviated health domain** (e.g. CHE) and sequential number of the indicator (e.g. CHE.IND.1, CHE.IND.2).

Component 8: High-level functional and non-functional requirements

- » Each functional requirement should have a **functional requirement ID** that should be structured as **abbreviated health domain** (e.g. CHE) and the sequential number of the functional requirement (REQ) (e.g. CHE.REQ.1, CHE.REQ.1).
- » Each non-functional requirement should have a **non-functional requirement ID** that should be structured as **abbreviated health domain** (e.g. CHE) and sequential number of non-functional requirements (NFXNREQ) (e.g. CHE.NFXNREQ.1, CHE.NFXNREQ.2).

DAK: digital adaptation kit; ID: identification.

How to use this digital adaptation kit

Target audience

The primary target audience for this DAK is health programme managers within the MOH, who will be working with their digital or health information systems counterparts in determining the health content requirements for a child health in humanitarian emergencies DTDS system. The health programme manager is responsible for overseeing and monitoring the implementation of the clinical practices and policies for the health programme area, in this case child health in humanitarian emergencies.

The DAK also equips individuals responsible for translating health system processes and guidance documents for use within digital systems with the necessary components to kick-start the process of developing a DTDS system in a standards-compliant manner. These individuals are also known as business analysts who interface between health content experts and software development teams. Specifically, the adaptation kit contains key outputs, such as the data dictionary and decision-support algorithms, to ensure the validity and consistency of the health content with the DTDS system.

Additionally, using this DAK requires a collaboration between health programme managers responsible for family planning and counterparts in digital health and health information systems. Although each DAK focuses on a particular health programme area (in this case child health in humanitarian emergencies), the DAKs are envisioned to be used in a modular format and link to other health programme areas within primary health care settings, in an effort to support integration across services.

Scenarios for using the digital adaptation kit

The DAK may be used across various scenarios, some of which are listed below.

Scenario 1

Incorporating WHO guideline content into existing DTDS systems

Countries that already have digital systems in place, such as electronic medical records and decision-support tools, may use the information in the DAK to cross-check whether the underlying content and data for specific health programme areas are aligned to WHO guidelines. Users of the DAK can identify and extract specific decision algorithms that would need to be incorporated into their existing digital systems. By reviewing this systematic documentation, health programme managers and implementers can more readily identify differences in workflows, data inputs and decision-support logics to examine the rationale for deviations and understanding local adaptations of guideline content.

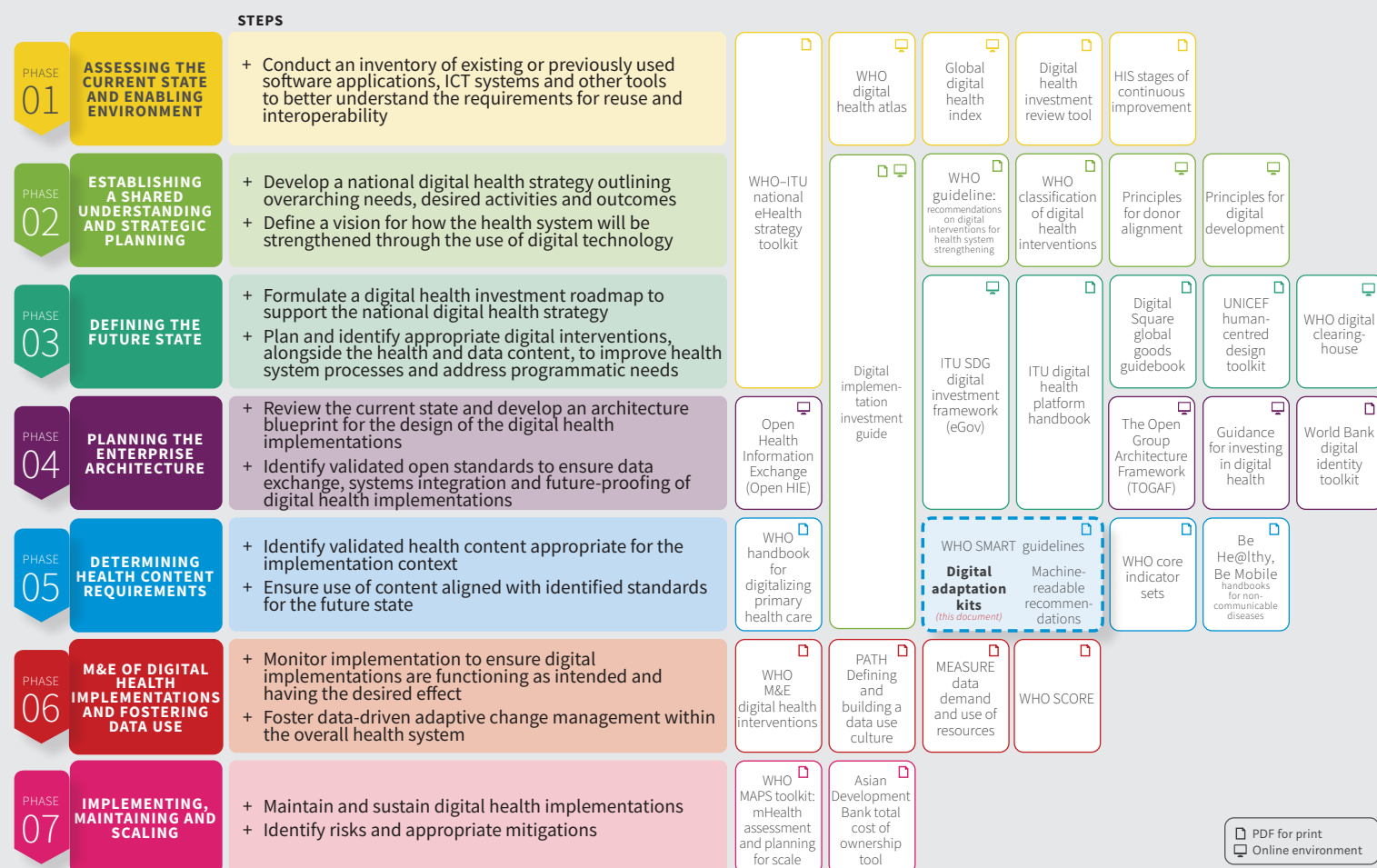
Scenario 2 Transitioning from paper to DTDS systems	Some countries may currently have paper-based systems that they would like to digitize. The process of optimizing paper-based client-level systems into digital records and decision-support may be overwhelming. Users in this scenario may review the DAK as a starting point for streamlining the necessary data elements and decision-support that should be in the optimized client-level digital system. Users may also then refer to the paper-based tools to determine whether there are missing fields or content that should also be included within the digital system. Users should also review the <i>WHO recommendations: optimizing health worker roles to improve access to key maternal and newborn health interventions through task shifting (OptimizeMNH) (14)</i>, which provides stepwise guidance on how to map data on paper-based forms into a digital system, including ways of accounting for data elements that are redundant or may not add value to the health system.
Scenario 3 Linking aggregate health management information system (HMIS) (e.g. District Health Information System 2 [DHIS2]) to DTDS systems used at point of care	In some instances, countries may already have a digital system for aggregate reporting and HMIS but may not yet have implemented digital systems that function at the service delivery level. The DAK can guide the development of a digital client record system that operates at point of care and ensure that there are linkages between the aggregate and service delivery levels (e.g. community or facility level). As such, a component of the DAK provides aggregate indicators derived from individual-level data to provide the linkage between these different levels. Indicators and performance metrics should also be consulted for supporting the use of routine data at the facility management and district levels.
Scenario 4 Leveraging data standards to promote interoperability and integrated systems	This DAK includes data elements mapped to ICD codes, and other standards, to support the design of interoperable systems. The data dictionary provides the necessary codes for different data elements, thus reducing the time for implementers to incorporate these global standards into the design of their digital systems. In addition, a critical part of service delivery in any health domain relies on engaging with clients. Digital interventions aimed at clients themselves, such as on-demand information services, targeted client communication (e.g. transmitting health information and reminders), reporting of health system feedback by clients on the quality of care, accessing their own medical records and home-based records, and self-monitoring of their health and diagnostic data (8), are all emerging approaches for complementing the services provided by health workers.

Links to the broader digital health ecosystem

DAKs represent one piece of the resources in the broader digital health ecosystem and should be used once there is a strategic vision by the MOH to use a DTDS system. In contexts where such vision may not exist, users should first consult the WHO–International Telecommunication Union *National e-Health strategy toolkit* (13), WHO *Recommendations on digital interventions for health system strengthening* (7) and the WHO *Digital implementation investment guide* (1) to establish a better understanding of how to select and apply appropriate digital health interventions. **Figure 3** situates DAKs within the broader set of resources for planning and implementing digital health systems.

Fig. 3

DAKs within the broader digital health ecosystem



ICT: information and communications technology; ITU: International Telecommunications Union; M&E: monitoring and evaluation; SMART: standards-based, machine-readable, adaptive, requirements-based and testable; WHO: World Health Organization.

Source: Adapted from WHO (1).

2

Part 2. Digital adaptation kit content for child health in humanitarian emergencies

Component



Health interventions and recommendations

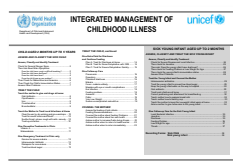
This DAK focuses on the following health interventions and recommendations related to child health in humanitarian emergencies.

1.1 Interventions based on WHO universal health coverage (UHC) list of essential interventions

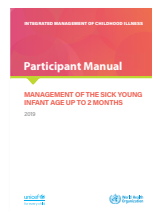
The key interventions for child health in humanitarian emergencies are not yet defined in the WHO UHC compendium of interventions (12). Therefore, these will be added in a future version of the DAK.

1.2 WHO guidelines, recommendations and guidance

The DAKs are intended to reflect health recommendations and content that has already been published in WHO guidelines and guidance documents. The health content and interventions for this DAK are drawn from the following WHO documents:



Integrated management of childhood illness (chart booklet) (March 2014) (15)



Management of the sick young infant aged up to 2 months: participant manual (16)

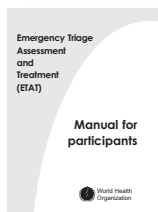


Integrated management of childhood illness (17)



Integrated management of childhood illness: distance learning course (18)

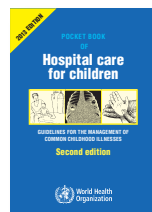
Other WHO guidance represented in the DAK include:



Emergency triage assessment and treatment (ETAT) (19)



Guideline on the prevention and management of wasting and nutritional oedema (acute malnutrition) (20)



Pocket book of hospital care for children: guidelines for the management of common childhood illnesses (21)



Integrated management of childhood illness: caring for newborns and children in the community (22)



Generic personas

A persona is a depiction of a relevant stakeholder, or end user, of the system. Although the specific roles and demographic profiles of the personas will vary depending on the setting, generic personas are based on WHO core competencies and credentials of different health worker personas. Please note that these are developed based on synthesis across multiple contexts as a starting point; further contextualization will be required according to the needs, motivations and challenges of the targeted personas in each setting.

2.1 Targeted generic personas

In the case of child health in humanitarian emergencies, health workers are the primary personas for the digital client health record and decision-support system. In the health systems surveyed for this DAK, the common combination of service providers was nursing professionals, nursing associate professionals, midwifery professionals, midwifery associate professionals, medical assistants, health care assistants, community health workers (CHWs) and doctors. A description of each persona is provided in [Table 2](#) (23).

A persona is a generic depiction of a relevant stakeholder or end user of the system. As specific roles, demographic profiles, challenges and motivations vary greatly depending on the context and setting, generic versions of a user persona should be included in the DAK. Thus, the generic personas for any health area should reflect WHO task shifting guidelines, national health workforce guidelines and general commonalities found across multiple settings. The creation of personas requires interviews with real people (i.e. health workers); there should be a distillation of the various interviews you have conducted into the generic personas. For example, if you have interviewed nurse midwives in two countries, a generic persona for a nurse midwife would be a combination of the commonalities between those two nurse midwives. Note that this is the area that will require significant contextualization into user personas in countries. Context-specific user personas can be added as an additional annex.

Table 2: Descriptions of key generic personas

Occupational title	Description	Different names	ISCO code (24)
Nursing professionals	Nursing professionals provide treatment, support and care services for people who need nursing care because of ageing, injury, illness or other physical or mental impairment, or potential risks to health, according to the practice and standards of modern nursing. They assume responsibility for the planning and management of the care of patients, including the supervision of other health workers, working autonomously or in teams with medical doctors and others in the practical application of preventive and curative measures in clinical settings.	Health worker Facility health worker	2221
Paramedical practitioners	Paramedical practitioners (including clinical officers and related) provide advisory, diagnostic, curative and preventive medical services more limited in scope and complexity than those carried out by medical doctors. They work autonomously or with limited supervision of medical doctors and perform clinical therapeutic and surgical procedures to treat and prevent diseases, injuries and other physical or mental impairments common to specific communities.	Clinical officer Primary care paramedic Advanced care paramedic Surgical technician Feldsher	2240
Nursing associate professionals	Nursing associate professionals provide basic nursing and personal care for people in need of such care because of the effects of ageing, illness, injury or other physical or mental impairment. They provide health advice to patients and families, monitor patients' conditions and implement care, treatment and referral plans usually established by medical, nursing and other health professionals.	Health worker Facility health worker	3221
Medical assistants	Medical assistants perform basic clinical and administrative tasks to support patient care under the direct supervision of a medical practitioner or other health professional. They perform routine tasks and procedures, such as measuring patients' vital signs, administering medications and injections, recording information in medical record-keeping systems, preparing and handling medical instruments and supplies, and collecting and preparing specimens of bodily fluids and tissues for laboratory testing.	Health worker Facility health worker	3256

ISCO: International Standard Classification of Occupations.

2.2 Related personas

In addition to the targeted personas detailed in [Table 2](#), there may be value in exploring other cadres and personas within the context of services for child health in humanitarian emergencies, such as the beneficiaries themselves (children and neonates), as well as medical secretaries, clerical support workers, and medical record and health information technicians. However, these were not identified as the central personas for the data and decision-support content detailed in this DAK. Additional related personas are listed in [Table 3](#).

Table 3: Descriptions of related personas

Name	Description	Different names (if relevant)	ISCO code (if relevant) (24)
Infant	Infants are less than 12 months old. A young infant refers to an infant younger than 2 months. They are the primary clients receiving health services from the targeted health worker personas. While this is a diverse population with different demographics and health needs, they generally have the following expectations: maintaining a healthy life for mother and baby (including preventing and treating risks, illness and death); maintaining physical and sociocultural normality; achieving positive infant health. The content specifications for infants and their carers will become even more important as additional client-side digital functionalities, such as targeted client communication (reminders), reporting of health system feedback and personal health tracking, are incorporated.	Infant/client/child	N/A
Child	A child is aged from 12 months up to 5 years. They are the primary clients receiving health services from the targeted health worker personas. While this is a diverse population group with different demographics and health needs, they generally have the following expectations: maintaining a healthy life (including preventing and treating risks, illness and death); maintaining physical and sociocultural normality; achieving positive childhood health. The content specifications for children and their carers will become even more important as additional client-side digital functionalities, such as targeted client communication (reminders), reporting of health system feedback and personal health tracking, are incorporated.	Child/client	N/A
Carer of infant/child	A person who gives care to an infant or child and can be considered the mother, father, adolescent mother, adolescent father, guardian or carer of the infant or child. They are the secondary clients receiving health services from the targeted health worker personas with respect to their infant or child. While this is a diverse population with different demographics and health needs, they generally have the following expectations: maintaining a healthy life for the infant or child (including preventing and treating risks, illness and death); maintaining physical and sociocultural normality for the infant or child; and achieving positive infant and child health. The content specifications for children and their carers will become even more important as additional client-side digital functionalities, such as targeted client communication (reminders), reporting of health system feedback and personal health tracking, are incorporated.	Caregiver	N/A
Medical secretaries	Medical secretaries, using specialized knowledge of medical terminology and health-care delivery procedures, perform several communication, documentation, administrative and internal coordination functions to support health workers in medical facilities and other health care-related organizations. They schedule medical appointments, record and review medical charts and correspondence, interview patients to complete case histories, prepare health insurance claims and acquisition orders, and supervise the work of other office support workers.	Facility staff	3344
Community health workers (CHWs)	CHWs provide health education, referral and follow-up, case management, and basic preventive health care and home visiting services to specific communities. They provide support and assistance to individuals and families in navigating the health and social services system.	Community health aide Community health promoter Village health worker Kaders (community volunteer)	3253

Name	Description	Different names (if relevant)	ISCO code (if relevant) (24)
Health care assistants	Health care assistants provide routine personal care, support and assistance with activities of daily living to patients and residents in a variety of health-care settings, such as hospitals, clinics and residential nursing care facilities. They assist patients with personal, physical mobility and therapeutic care needs as per established care plans and practices, and generally under the direct supervision of medical, nursing or other health professionals or associate professionals.	Nursing aide (hospital or clinic) Patient care assistant Birth assistant (hospital or clinic) Psychiatric aide	5321
Pharmacists	Pharmacists store, preserve, compound and dispense medicinal products. They counsel on the proper use and adverse effects of drugs and medicines after prescriptions are issued by medical doctors and other health professionals. They contribute to researching, testing, preparing, prescribing and monitoring medicinal therapies for optimizing human health.	Hospital pharmacist Industrial pharmacist Retail pharmacist Dispensing chemist	2262
Clerical support workers	This group covers clerical support workers (excluding specialized health information technicians and medical secretaries) working in health systems, including, for instance, general clerks, keyboard clerks, client services clerks, material recording clerks and others who record, organize, store, compute and retrieve information, and perform clerical and secretarial duties in connection with money-handling operations, requests for information and appointments.	Clerk	N/A
Health management personnel not classified elsewhere	This group covers managers and administrators not classified elsewhere (except health service managers) working in health systems, including, for instance, government health department heads, human resource managers, supply chain managers, regional health policy and planning directors, and others whose main tasks and duties include guiding and directing the activities of organizations, departments and other workers.	District health officer	N/A
Medical records and health information technicians	Medical records and health information technicians develop, implement and assess health records processing, storage and retrieval systems in medical facilities and other health-care settings to meet the legal, professional, ethical and administrative record-keeping requirements of health services delivery.	Facility administrator	3252
Other health service providers not classified elsewhere	This group covers other categories not classifiable as participating in the formal or informal health labour market but providing health services, including, for instance, medical interns and trainees who are providing clinical services as part of their basic medical education.	Medical student intern Hospital volunteer	

ISCO: International Standard Classification of Occupations; N/A: not applicable.

2.3 Additional considerations for contextualizing personas

Although this section provides an overview of the generic roles of the targeted personas, it is important to contextualize these personas to your setting. The generic and related personas described in [Tables 2](#) and [3](#) can be supplemented by reflecting on these additional considerations:

- » **background and demographics** (e.g. sex; age; whether they are from the community; familiarity with digital devices; do they own a mobile phone or smartphone?);
- » **local environment** and any relevant contextual information about their surroundings (e.g. work site characteristics; rural or urban; availability of electricity, water, internet; distance from the nearest referral facility);
- » **expected roles and responsibilities** (What are the expected roles and responsibilities based on country context? How does this differ from the roles and responsibilities defined by WHO?);
- » **actual roles and responsibilities** (What are their actual roles and responsibilities, if there is any difference from what is expected?);
- » **context** (What is the level of internet connectivity? How are they compensated? How far away is the nearest referral facility? What other personas or health workers do they interact with?);
- » **challenges** (What are the day-to-day challenges the end user might face?); and
- » **motivations** (What does success look like to them? Are there targets they need to achieve?).

For more details on persona development, please refer to the *WHO recommendations: optimizing health worker roles to improve access to key maternal and newborn health interventions through task shifting (OptimizeMNH) (14)*.

Component



User scenarios

3.1 How to interpret user scenarios for functional requirements

User scenarios are helpful tools not only to better understand the context in which a digital tool would operate, but also for some insights into what key data elements would need to be recorded and accounted for in the database. Additionally, the context in which the tool would operate, illuminated by the user scenarios, provides an insight into some functional and non-functional requirements that the system would also need. Highlighted in **yellow** are some key data elements that need to be recorded or calculated. Highlighted in **blue** is some decision-support logic that can be automated in the system. Highlighted in **green** are some key functional and non-functional requirements that should be included in the system; in **bold** are some adolescent-specific considerations that should be accounted for. For example, the interpretation of the user scenarios for first contact/new client, existing child, sick child, and first contact/new client, and sick young infant is shown in [Tables 4–6](#).

Table 4: Interpretation of the user scenario 1: first contact/new client, sick child

Data element to be collected	Decision logic to be embedded	Functional and non-functional requirements
Unique ID	Classify the child's health problem	Search for the patient
Name	Determine the treatment to be given	Register a new child at the facility with a unique ID
Date of birth	Determine health promotion and immunization needs	Calculate fast breathing and weight-for-age using inputs
Weight		Pop-up appears telling the health worker what certain signs looks like in a child and provides an image
Type of visit		Save and close the consultation
Danger signs		
Symptoms – cough		
Respiratory rate		

Table 5: Interpretation of the user scenario 2: existing client, sick child

Data element to be collected	Decision logic to be embedded	Functional and non-functional requirements
Unique ID Weight Type of visit Danger signs Symptoms – cough Duration of cough Respiratory rate Oxygen saturation	Classify the child's health problem Determine the treatment to be given Determine health promotion and immunization needs	Search for the patient Find the existing patient and start a new consultation Calculate fast breathing, for example, weight-for-age using inputs Pop-up appears telling the health worker what certain signs looks like in a child and provides an image Save and close the consultation

Table 6: Interpretation of the user scenario 3: first contact/new client, sick young infant

Data element to be collected	Decision logic to be embedded	Functional and non-functional requirements
Unique ID Name Date of birth Sex Caregiver name Caregiver phone number Caregiver relationship Temperature Weight Danger signs Symptoms	Classify the child's health problem Breastfeeding assessment results Determine the treatment to be given Determine health promotion and immunization needs	Register a new child at a facility with a unique ID Calculate weight-for-age using z-scores Save and close the consultation

3.2 User scenario for scenario 1: first contact/new client, sick child

Key personas	Mother: <i>Amira</i>	Child: <i>Jacob</i>	Health worker
Amira arrives with her infant Jacob, who is 6 months old, at the clinic. Amira tells the health worker that Jacob is not feeling well and she is worried about him. This is the first time that Amira has visited this clinic with Jacob and it is the first visit to a health-care facility for Jacob's illness. The health worker creates a new client record for Jacob using the digital tool. This includes entering demographic information about Jacob, notably his name, date of birth and sex. The health worker reviews Jacob's child health card, asks Amira to confirm the information and enters it into the tool. The tool prompts the health worker to collect information about Jacob's caregivers. The health worker asks Amira about her and Jacob's father, including their names and how they can be contacted. The health worker enters this information into the tool and saves it to Jacob's client profile so that it will be available each time he visits the clinic. Then, the health worker is ready to enter new information into the tool about this consultation with Jacob. The tool prompts the health worker to select the type of visit for this consultation; as Jacob is sick, the health worker selects the sick child visit option. Amira tells the health worker that she has noticed for the last 3 days that Jacob does not seem to be himself. He has started on some solid foods but he is now not eating as much as he was. He is also not breastfeeding as much as he was and has a cough. The digital tool prompts the health worker to assess Jacob for danger signs of severe illness. The health worker concludes that Jacob has no danger signs and enters this information into the tool. The tool then prompts the health worker to take basic measurements of Jacob. The health worker weighs Jacob and takes his temperature. Jacob weighs 5 kg and his temperature is 37 °C. The health worker measures Jacob's mid-upper arm circumference (MUAC) and height and enters the information gathered into the tool. The tool indicates to the health worker that Jacob's temperature is normal. The tool calculates Jacob's weight-for-age and weight-for-height z-scores based on his age and the information the health worker has entered and tells the health worker that both are normal. After the health worker has entered the information about the measurements into the tool, the tool prompts the health worker to assess for symptoms of illness or problems. The health worker asks Amira about all the symptoms she has noticed and Amira tells the health worker that she has noticed that Jacob starts to breastfeed and then pulls off the breast and seems to pant for air before going back for a further suckle. She has noticed that his stomach is going in and out quite quickly when he breathes and he has a cough. The tool prompts the health worker to ask Amira specific questions about the symptoms Jacob has been experiencing. Specifically, if he has had a cough or difficulty breathing, diarrhoea, a history of fever or an ear or skin problem. The health worker enters in the tool that Jacob has difficulty breathing and a cough; all other symptoms are not present or are normal. When the health worker records in the tool that Jacob has a cough, the tool prompts the health worker to ask the caregiver for how long Jacob has had a cough. The health worker asks Amira and she reports that Jacob has been coughing for 3 days. The health worker records this in the tool. The health worker is then prompted by the tool to check Jacob's illness prevention history, including receipt of vitamin A, deworming status and immunizations. The health worker reviews Jacob's child health card and asks Amira about when his last health promotion services were; the health worker notices that Jacob has not received vitamin A and is due for immunization.	Corresponding business processes (see Component 4)	The health worker is then prompted to look for signs of illness and carry out a physical examination. The health worker is prompted to see if Jacob has chest indrawing when he breathes. The health worker does not remember exactly what chest indrawing looks like, so they click on the information button in the tool. A description pops up explaining what chest indrawing looks like in a child and provides an image to explain this sign to the health worker. After reviewing the information, the health worker decides Jacob does not have chest indrawing. The health worker then measures Jacob's respiratory rate and counts 60 breaths per minute. The health worker enters this into the tool; a message pops up specifying that based on his respiratory rate and age, Jacob has fast breathing. For his age, 50 breaths or more is considered fast. The health worker completes his physical examination of Jacob and enters the findings into the tool as they conduct the exam. All other signs are normal or not present. No further assessments, tests or laboratory tests are prompted by the tool for Jacob. Now that the assessment is complete, the tool suggest a classification of illness. The tool suggest that Jacob has pneumonia, no acute malnutrition and no anaemia. It also suggests that Jacob meets the criteria for prevention, screening and other problems, and that vaccination and vitamin A are needed. The health worker is given the option to agree or disagree with each of the classifications proposed by the tool. The health worker agrees with all the proposed classifications. For each of Jacob's classifications, the tool proposes recommended treatments and management. The tool suggests that for pneumonia, Jacob should be treated with oral amoxicillin; it provides the appropriate dose, number of doses per day and duration of treatment for Jacob, based on his age, weight and illness classification. The tool reminds the health worker that Jacob should be followed up in the clinic in 3 days for pneumonia. Because Jacob has not received vitamin A, the tool reminds the health worker that Jacob is eligible for vitamin A at this visit and reminds the health worker to conduct any vaccinations for which Jacob is eligible. The tool prompts the health worker to assess Jacob for any feeding problems and reminds the health worker of the counselling that should be provided to Amira to care for Jacob at home. This includes a safe remedy for soothing Jacob's cough, feeding recommendations, how to administer the medication Jacob has been prescribed and when to return. The health worker counsels Amira on how to treat Jacob's pneumonia and how she should give the antibiotic and the throat remedy at home, and asks her to bring Jacob back in 3 days to check on his condition. The health worker reviews with Amira the signs that she should be aware of and explains that if Jacob experiences these signs, she should immediately return to the clinic with him. The health worker saves Jacob's record in the digital tool.	This scenario refers to the following business processes: - Register the child - Access and classify the child - Treat the child

3.3 User scenario for scenario 2: existing client, sick child

Key personas	Mother : Aisha Child: Abbas Health worker
Aisha is a 23-year-old woman who has brought her two and a half year old son, Abbas, into a primary health care facility because Abbas has been sick for the last few days. The health worker asks Aisha if this is Abbas's first visit to this facility. Aisha says she took Abbas to this facility before when he was younger and had a different problem. The health worker asks the reason for today's visit and Aisha reports that Abbas has had a cough for a few days. The health worker looks for Abbas in the registered patients list in the tool, using his name and date of birth, and finds his profile. Abbas's profile already includes his basic demographic information, as well as information about Aisha and Abbas's father. The health worker selects Abbas's profile and starts a new consultation for Abbas. The health worker now checks for any danger signs of severe illness that Abbas may be experiencing, as prompted by the tool. Abbas does not have any danger signs, so they proceed with the consultation. The health worker measures Abbas's temperature, weight, MUAC and height as prompted by the tool. The tool calculates the weight-for-height z-score and the weight-for-age z-score; both are normal (z-score = 0). The health worker is now prompted by the tool to check for symptoms of illness. Abbas has a cough, and when the health worker enters this into the tool, the tool suggests an additional question regarding the duration of the cough. The health worker asks Aisha how long Abbas has been coughing, and enters 4 days into the tool, based on Aisha's response. The rest of the symptoms are absent or normal. The next questions relate to the prevention and health promotion services Abbas has received. The health worker asks Aisha for her child's health record to determine his history of HIV counselling and testing, vaccination status, last doses of vitamin A and deworming. The health worker now looks for signs of illness by conducting a physical examination. The tool reminds the health worker to evaluate specific signs based on Abbas's age and his symptoms. These include respiratory rate and oxygen saturation. The health worker observes that Abbas is wheezing when he breathes. When the health worker measures his oxygen saturation, it is 89%. When the health worker enters this into the tool, a pop-up highlights that Abbas's oxygen saturation is very low and reminds the health worker to make sure that the measurement was performed correctly, to give oxygen if available and to quickly complete the assessment not to delay referral of the child. Once all the data and signs are entered in the system, the tool suggests that Abbas has severe pneumonia with wheezing. Based on what the health worker has observed, they agree with this diagnosis. Pre-referral for treatment using intramuscular ampicillin and gentamicin and a fast-acting bronchodilator is suggested. The health worker arranges for Abbas to be referred. The health worker explains to Aisha that they believe Abbas has severe pneumonia and should be referred to the hospital for further treatment. He asks Aisha if she has any questions or concerns. The health worker saves the information about Abbas's visit in the tool.	
Corresponding business processes (see Component 4)	This scenario refers to the following business processes: A. Search for the existing child B. Access and classify the child C. Treat the child

3.4 User scenario for scenario 3: first contact/new client, sick young infant

Key Personas

Mother: Radha
Child: Savi
Health worker

Radha arrives at the facility with her young infant Savi. Radha tells the health worker that Savi's skin seems a bit yellow and she is very worried.

This is her first visit to the clinic since Savi was born. The health worker registers Savi in the tool by entering her name, date of birth, sex and other demographic information. The health worker saves Savi's profile in the tool, which will include information about her and her caregivers for use on later visits. Then, the health worker opens a new consultation for Savi. Because Savi has a problem, the health worker selects sick infant/child as the visit type.

First, the tool prompts the health worker to evaluate whether Savi has any signs of a life-threatening illness. The signs the tool prompts depend on Savi's age. After recording in the tool that Savi has no signs of a life-threatening illness, the health worker collects basic measurements for Savi, including weighing her and taking her temperature. When the health worker enters Savi's temperature into the tool, the tool prompts them to tell Radha that the temperature is normal for a baby Savi's age. When the health worker enters Savi's weight, the tool calculates the weight-for-age z-score and the alerts the health worker that Savi's weight-for-age z-score is low.

Next, the health worker asks Radha about Savi's problems. Radha explains that she noticed that Savi had yellowish skin and this was different from her other babies when they were little. The health worker is prompted by the tool to ask about the symptoms of possible illnesses or problems that Savi may be experiencing, including whether she had had convulsions in this illness, how she is feeding, whether she has diarrhoea and whether she is breastfed. The health worker discusses this with Radha and records in the tool that none of the symptoms of illness are present, and the baby is breastfed.

Next, the health worker assesses what health promotion and screening services Savi has recently received, including the HIV counselling and testing history of Radha and Savi's vaccination history. The health worker reviews Savi's child health card and asks Radha to confirm that Radha was tested for HIV in pregnancy and had a negative result. Savi received vaccinations at the hospital when she was born. The health worker enters this into the tool.

Next, the health worker looks for signs of illness in Savi by conducting a physical examination. They observe Savi's breathing and ensure there is no chest indrawing, count the breaths Savi takes in one minute and observe how she is moving. They look at her skin and umbilicus to make sure there is no sign of infection and look at the colour of Savi's skin. When the health worker enters into the tool that Savi has yellow skin, the tool prompts a question as to when jaundice first appeared. The health worker asks Radha, who says the yellow skin appeared yesterday, when the baby was 3 days old, and the health care worker enters in the tool that it began after the first 24 hours of life.

The health worker then has a discussion with Radha about how she feeds Savi, to learn more about her breastfeeding practices, and enters this information into the tool. The health worker then conducts an observation of Radha breastfeeding Savi to ensure there are no problems with feeding. The health worker does not see any breastfeeding problems and enters this into the tool.

Now that the health worker has finished the look, listen and feel components of the consultation, the tool provides a list of proposed classifications for Savi: infection unlikely, jaundice, feeding problem or low-weight-for-age, weight-for-age, HIV infection unlikely. The health worker reviews each of these classifications, thinks about what they have observed in the consultation and decides if they agree with them. The health worker is then provided with a list of treatments and management options for each of the classifications. This includes counselling on appropriate breastfeeding, how to keep her baby warm and how to care for her baby at home. The health worker explains to Radha that her baby has jaundice and reassures her that Savi can be cared for at home. The health worker asks Radha to return with Savi tomorrow to check on the condition of the jaundice, and in 14 days (about 2 weeks) to check Savi's weight. The health worker explains feeding recommendations, how to keep the baby warm at home and reminds Radha when to return immediately with the baby, including if the palms of Savi's hands or the sole of her feet are yellow. Then, the health worker saves the information about the consultation with Savi in the tool.

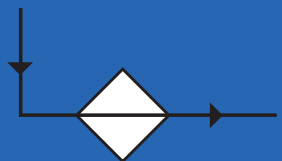
Corresponding business processes
 (see Component 4)

This scenario refers to the following business processes:

- A. Register the child
- B. Access and classify the child
- C. Treat the child

Component

4



Business processes and workflows

A business process, or process, is a set of related activities or tasks performed together to achieve the objectives of the health programme area, such as registration, counselling and referrals (1, 13). Workflows are a visual representation of the progression of activities (tasks, events, interactions) that are performed within the business process (13). The workflow provides a story for the business process being diagrammed and is used to enhance communication and collaboration among users, stakeholders and engineers.

This DAK focuses on key business processes conducted by the health worker persona within primary health care facility service provision. These business processes are described in Table 7. For each of these business processes, the corresponding business processes, data elements and decision-support needs are detailed within the following sections of this document.

Table 7: Overview of key child health in humanitarian emergencies module business processes^a

No.	Process name	Process ID	Personas	Objectives	Task set
	Title	ID used to reference this process throughout the DAK	Individuals interacting to complete the process	A concrete statement describing what the process seeks to achieve	The general set of activities performed within the process
A	Registration	CHE.A. Registration	» Child » Caregiver » Health worker	This process captures the personal details of the patient and creates a new registration record. Child details such as name, age, and the caregiver details are captured.	Starting point: the sick/well child arrives at the facility: » gather the child details; » review and update the child record; » create a new child record.
B	Access and classify the sick child	CHE.B. Access and classify the sick child	» Child » Caregiver » Health worker	Check for danger signs, symptoms and signs, perform the required tests and record the observations. Based on the observations, use the system to classify the sick child. In case of well child, it is about recording and managing the immunizations and other preventive measures.	Starting point: the child is registered at the facility (check for danger signs): » complete the basic measurements; » access and examine the symptoms and signs; » complete the required assessments and record the test results; » confirm or agree with the classifications proposed by the system.
C	Treat the child	CHE.C. Treat the child	» Child » Caregiver » Health worker	Provide the drug treatment, other treatment and advice to the caregiver to list the next steps for follow-up or referral.	Starting point: the child has been classified and confirmed by the health worker: » display the main treatment, other treatments and advice for the caregiver; » determine the follow-up or referral as part of next steps.

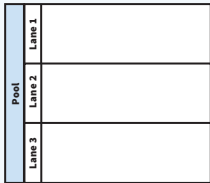
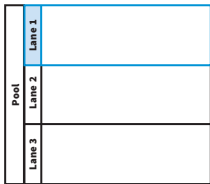
^a The editable files of each business process in .bpmn format can be found [here](#).

Processes that are a part of child health in humanitarian emergencies service delivery but are not included in this DAK include: configuration; billing; dispensing; and how to follow up (although follow-up visits are part of the child health emergency counselling workflow). These processes may be required, although this is highly country-specific and context-specific. If applicable, the business process could include an insurance or coverage check, which could take place at any one of many points during an emergency care visit.

4.1 Overview of key processes

This section illustrates the workflows of the identified processes using standardized notations for business process mapping. [Table 8](#) first provides an overview of these notations.

Table 8: Notations used in business process mapping

Symbol	Symbol name	Description
	Pool	A pool consists of multiple “swim lanes” that depict all the individuals or types of users involved in carrying out the business process or workflow. Diagrams should be clear, neat and easy for all viewers to understand the relationship across the different swim lanes. For example, a pool would depict the business process of conducting an outreach activity, which involves multiple stakeholders represented by different lanes in that pool.
	Swim lane	Each individual or type of user is assigned to a swim lane , a designated area for noting the activities performed or expected by that specific actor. For example, a health worker may have one swim lane; the supervisor would be in another swim lane; and the clients or patients would be classified in another swim lane.
	Start event or trigger event	The workflow diagram should contain both a start and an end event , defining the beginning and completion of the task, respectively.
	Start event message	The flow starts with a message . The meaning of “message” in business process model and notation is not restricted to letters, emails or calls. Any action that refers to a specific addressee and represents or contains information for the addressee is a message.
	End event	There can be multiple end events depicted across multiple swim lanes in a business process diagram. However, for diagram clarity, there should only be one end event per swim lane.
	Activity, process, step or task	Each activity should start with a verb, for example, “register client”, “calculate risk”. Between the start and end of a workflow, there should be a series of activities noting the successive actions performed by the actor for that swim lane. There can also be subprocesses of each activity.
	Activity with subprocess	This denotes an activity that has a much longer subprocess to be detailed in another diagram. If the diagram starts to become too complex and unhelpful, the subprocess symbol should be used to reference another process depicted on another page.

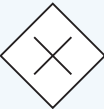
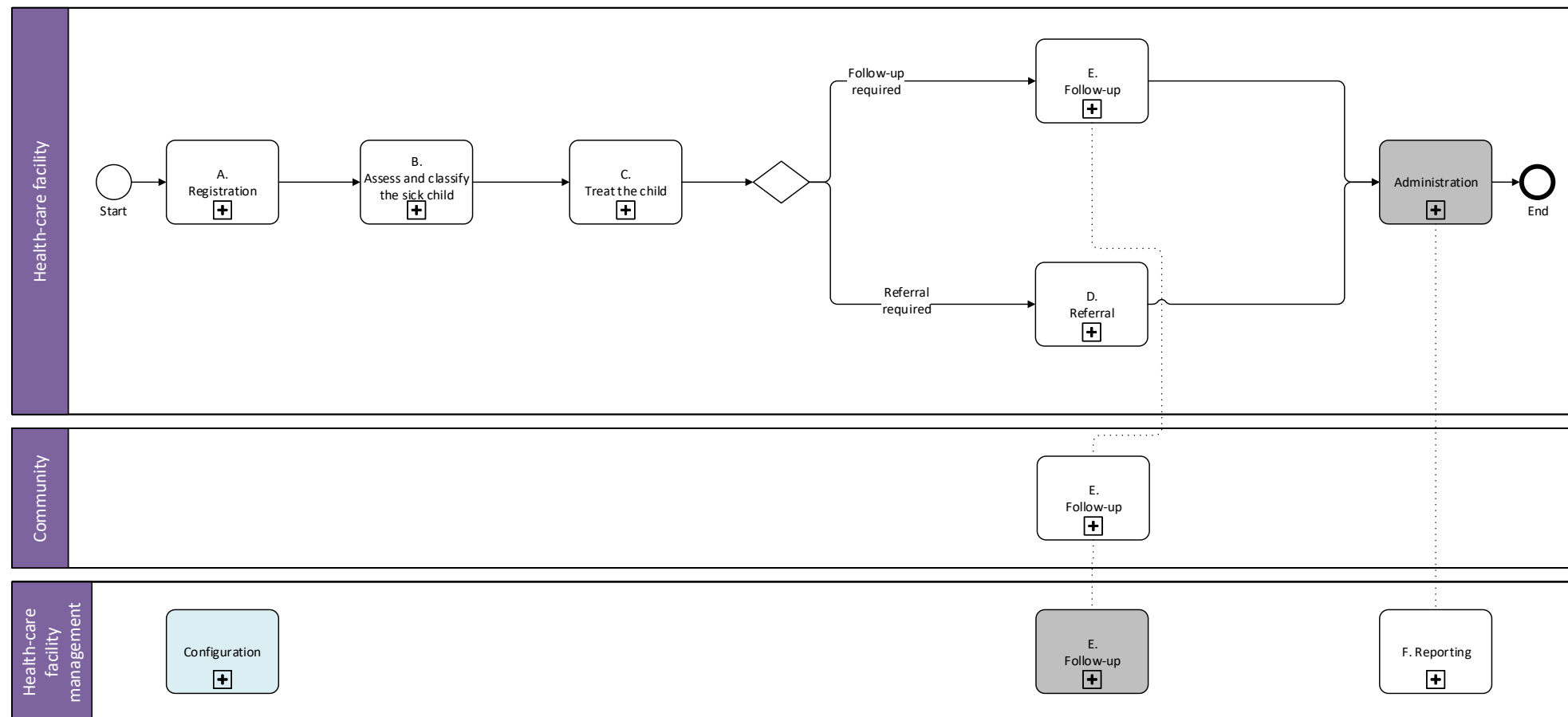
Symbol	Symbol name	Description
	Activity with business rule	This denotes a decision-making activity that requires the business rule , or decision-support logic, to be detailed in a decision-support table. This means that the logic described in the decision-support table will come into play during this activity, as outlined in the business process. This is usually reserved for complex decisions.
	Sequence flow	This denotes the flow direction from one process to the next. The end event should not have any output arrows. All symbols (except for start event) may have an unlimited number of input arrows. All symbols (except for end event and gateway) should have one and only one output arrow, leading to a new symbol, looping back to a previously used symbol or to the end event symbol. Connecting arrows should not intersect (cross) each other.
	Message flow	This denotes the flow of data or information from one process to another. This is usually used for when data are shared across swim lanes or stakeholder groups.
	Exclusive gateway	This symbol is used to depict a fork, or decision point, in the workflow, which may be a simple binary (e.g. yes/no) filter with two corresponding output arrows, or a different set of outputs. There should only be two different outputs that originate from the decision point. If you find yourself needing more than two output or sequence flow arrows, you most likely are trying to depict decision-support logic or a business rule. This should be depicted as an activity with business rule (above) instead.
	Parallel gateway	The parallel gateway can be used to model concurrency in a process. This type of gateway allows forking into multiple paths of execution or joining multiple incoming paths of execution. An important difference with other gateway types is that the parallel gateway does not evaluate conditions.
	Terminate end event	A terminate end event indicates that all activities in the process have to be immediately ended, including all the instances of multi-instance activities. The process will be ended without any compensation or event handling. It marks the fact that the business process terminated without reaching the intended end.
	Throw – link event	The throw – link event serves as the start of an off-page connector. It is the end of the process when there is no more room on your page for that workflow. It is the end of a process on your current page or the end of a subprocess that is part of a larger process. A catch – link will need to follow the throw – link .
	Catch – link event	The catch – link serves as the end of an off-page connector. It is the start of the new process on a different page from the throw – link or the start of a subprocess that is part of a larger process. A throw – link must be aligned to the catch – link .
	Annotations	Annotations are used to supplement the diagrams. Annotations contain useful (additional) information and can be connected to other elements with associations.
	Ad hoc marker	A container with an ad hoc marker can contain multiple activities (tasks or subprocesses), which can be executed in any order, executed several times, or skipped. However, not all these activities need to be finished before moving on to the next activity.
	Loop marker	A container with loop marker indicates that the activities inside the container repeat until a defined condition either applies or ceases to apply. The condition on which a loop executes can be linked to the container as an annotation.

Figure 4: Overview of key child health in humanitarian emergencies processes



For the key, see [Table 8](#).

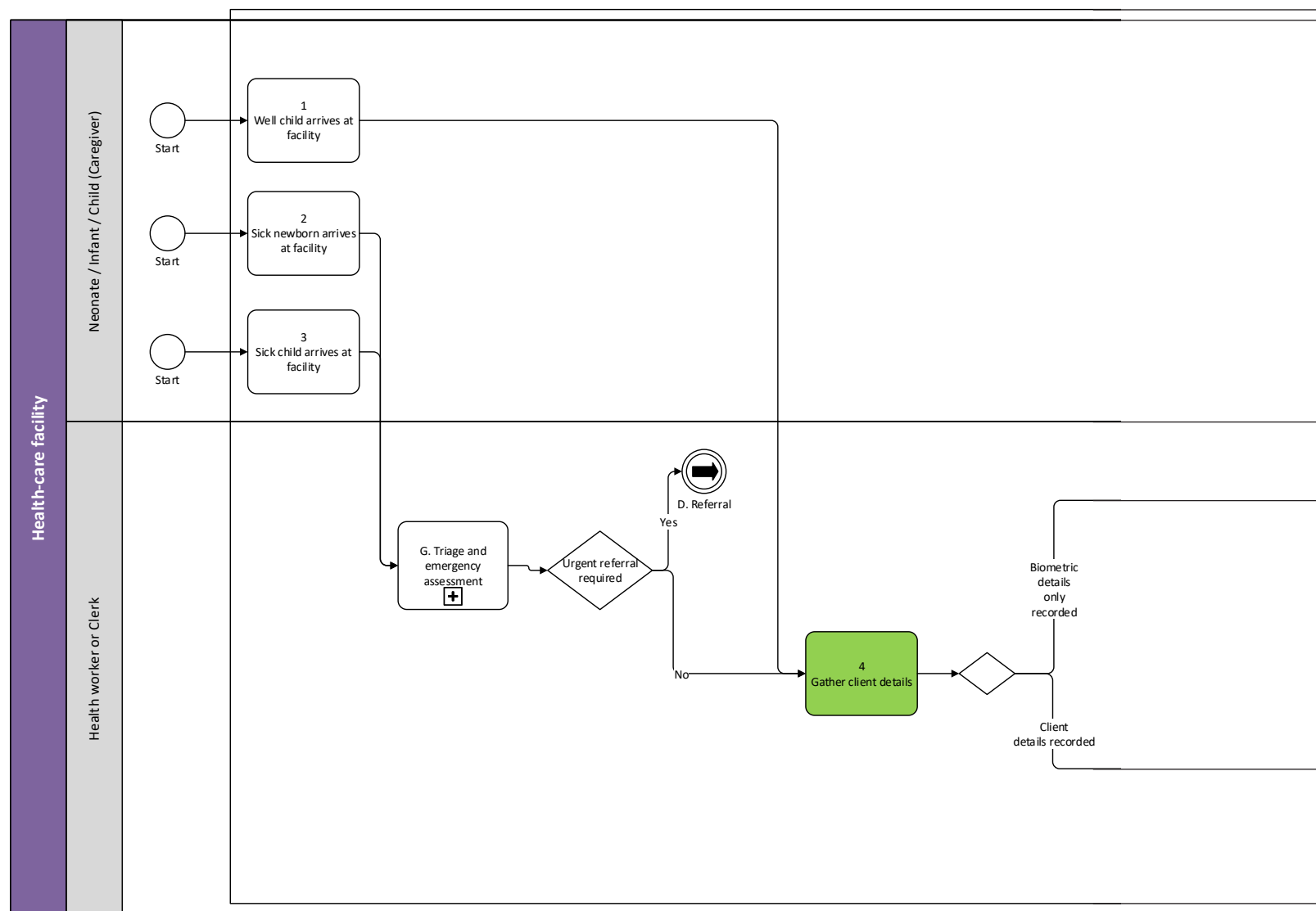
4.2 Workflows

Workflows represent the progression of activities performed within the business process. They help users and stakeholders understand the relationship between activities, data elements and decision-support needs. The workflows shown in this section depict processes that have been generalized and may not reflect variation and nuances across different settings. Also, the simplicity of the workflow may not adequately illustrate non-linear steps that may occur. Even though in some workflows counselling and obtaining informed consent activities are illustrated and described only in one activity, those actions occur throughout the entire process.

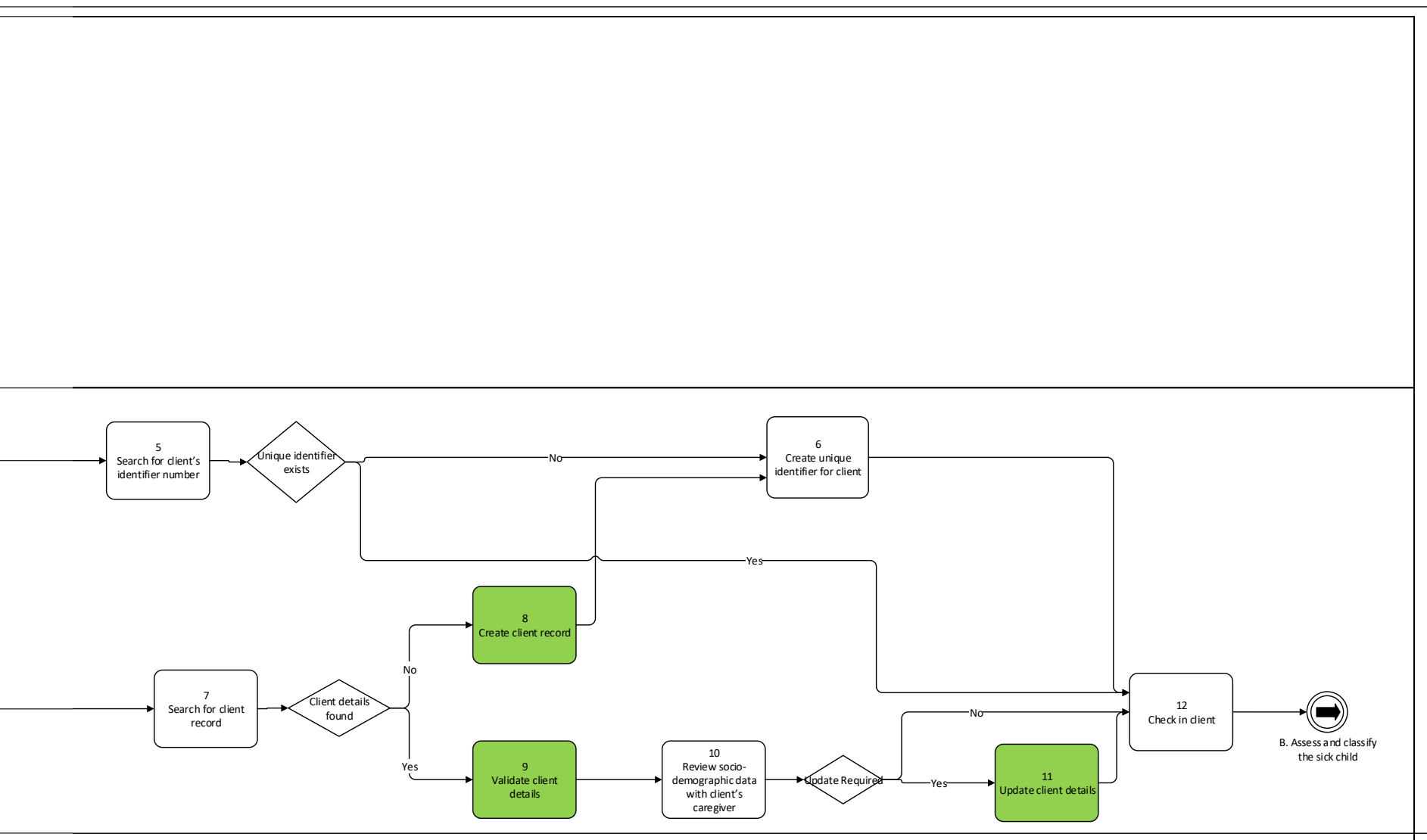
A. Business process for registration

Objective: To ensure that child details such as name and age, and caregiver details are captured and stored in the system, and that the child is ready for consultation (Fig. 5).

Figure 5 Workflow A: registration business process



For the key, see [Table 8](#).
Source: Adapted from *Health Level 7 (25)*.



REGISTRATION BUSINESS PROCESS NOTES AND ANNOTATIONS

General note

Registration may be conducted as a stand-alone process by a data clerk or administrative persona ahead of the child consultation or it may be conducted directly by the health worker as part of the consultation.

1. Arrive at the facility

- » The caregiver arrives with the child at the facility.
- » The child could be already registered at the facility, and it could be a repeat visit for them.

2. Gather the client details

- » Ask the caregiver whether they have previously been issued a unique identifier for their child.
- » Does the caregiver have a card, number or barcode?
- » Do they say that they are a returning or a referred patient?
- » If a referral, check for a referral slip or data from the community.
- » Determine whether the child is new to the health-care facility or health post.
- » For returning clients, details will be retrieved from the registry of children at this facility or, if possible, from a central patient registry.

3. Search for the client

- » This search process can be done through a variety of means depending on what mechanisms are available in the country. For example, patients can be searched for by using their name, unique identifier or client record.
- » If the patient is not found, capture the child details and create a new record.
- » Issue a unique identifier, if used and possible at the facility.

4. Validate the client details

- » Review and update the patient record.
 - **4.1. Review the sociodemographic data with caregiver:**
 - Review the child's non-clinical information, such as name, age and caregiver contact information.
 - **4.2. Is an update needed?**
 - Have they changed their contact information or has any other sociodemographic information changed?
 - **4.3. Update the client details**
 - The caregiver can provide updated information if they have changed their details recently.
- » Merge or update the patient record.
- » This may also happen during the counselling process.

5. Check in the client

- » Record the child's updated details in the patient registry.

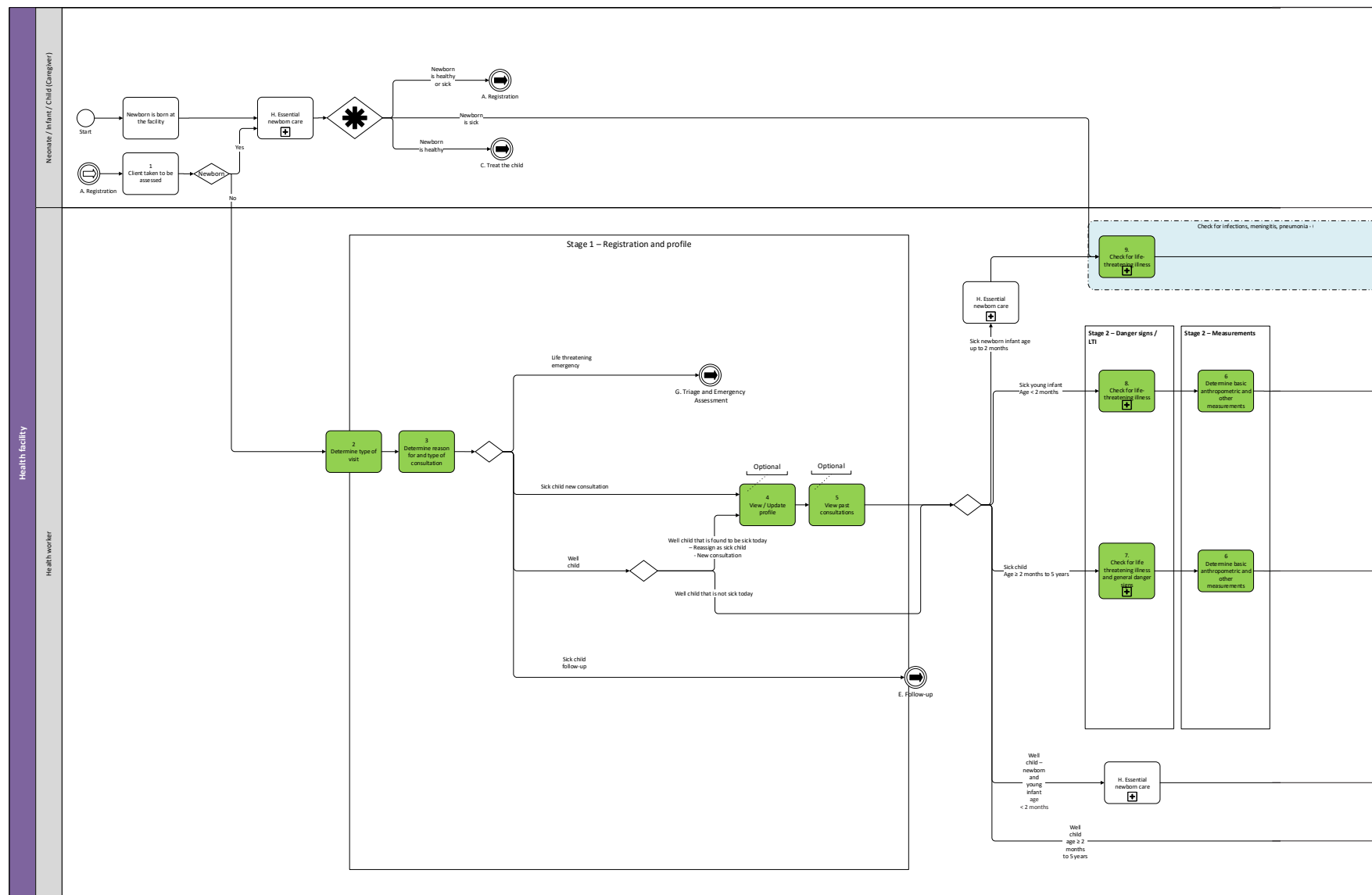
6. Counselling

- » After a child is registered, move on to the access and classify workflow.

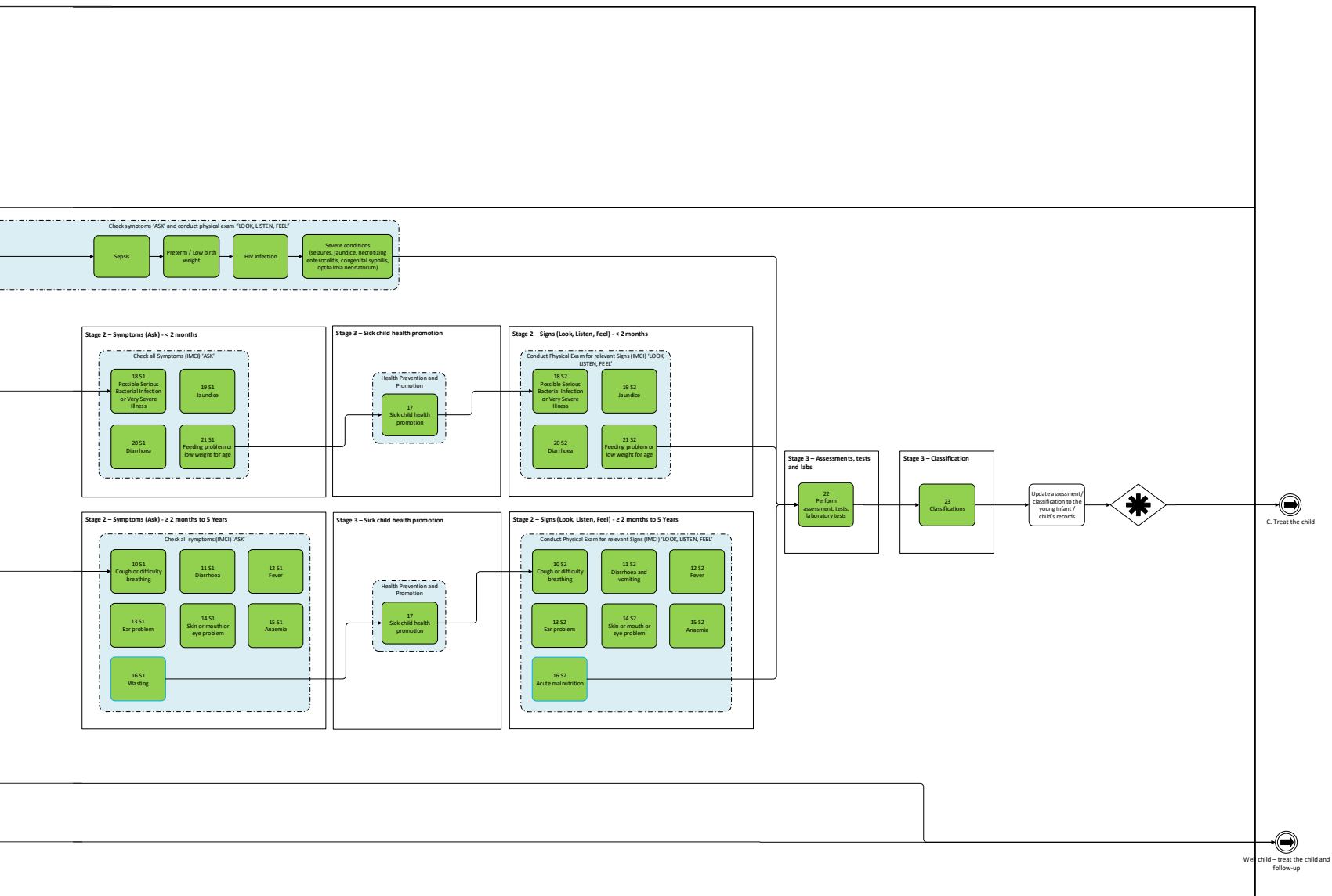
B. Business process for assess and classify

Objective: To check for danger signs and symptoms, perform the required tests and record the observations. Based on the observations, use the system to classify the sick child. If the child is well, it is a matter of recording and managing the immunizations and other preventive measures (Fig. 6).

Figure 6 Workflow B: assess and classify business process



For the key, see [Table 8](#).



ASSESS AND CLASSIFY BUSINESS PROCESS NOTES AND ANNOTATIONS

General notes

Check for the type of visit and reason for the consultation. Depending on whether the child is sick or well, different subprocesses should be followed.

1. Determine the reason for the visit

- » Search for the child details in the system and determine the reason for the visit.

2. Life-threatening illness or emergency: check if the child is showing any danger signs or signs of a life-threatening illness

- » Stabilize the child and then refer them and continue the consultation.

3. Capture or update the child history

- » Discuss and review the available records for the child.
- » Update the child profile if required.

4. Basic anthropometric and other measurements

- » Measure the child's temperature, weight, height and MUAC.

5. Symptoms and signs: check the symptoms as per the Integrated Management of Childhood Illness (IMCI) and ask the caregiver relevant questions to get all the information

- » Look and feel for the signs depending on the symptoms that are discussed. Conduct a physical examination whenever required and note the observations.

6. Health promotion and assessments

- » Check for vitamin A status, deworming and immunizations as relevant to the age of the child and record the details.
- » Perform the assessments and required laboratory tests as suggested by the system. Enter the observations and test results in the system.

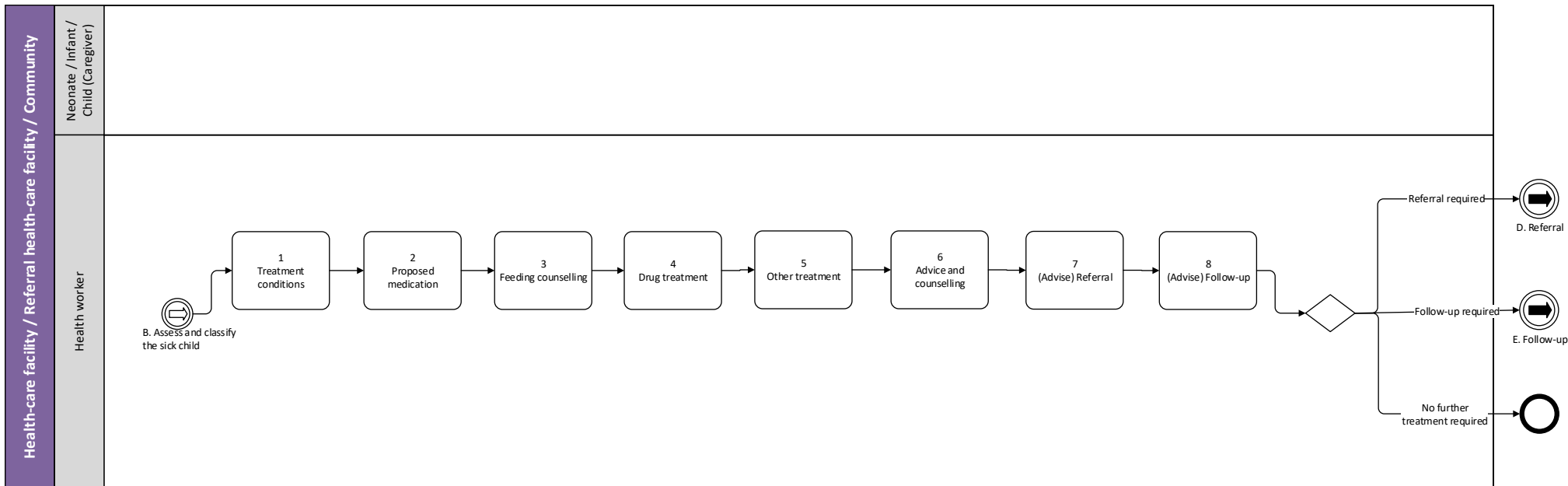
7. Classifications and treatments

- » Based on the decision logic, classifications will be prompted. Agree or disagree based on the health worker observations. Treatments, advice and the next follow-up steps will be prompted by the system. Store the classification and treatment details as part of the child's profile.

C. Business process for treating the child

Objective: To provide the caregiver with drug treatment, other treatments and advice, and list the next steps for follow-up or referral (Fig. 7).

Figure 7 Workflow C: treat the child business process



For key, see Table 8.

TREATMENT STAGE NOTES AND ANNOTATIONS

1. Proposed medication and drug treatment

- » For the specific classification, with or without qualifiers and additional input data, the system proposes the medication and drug treatment.
- » The health worker should follow the treatment suggestions and provide treatment to the child. The health-care facility reviews the accuracy, validity and completeness of the data in the system.
- » This can be automated and done digitally.

2. Other treatment and advice to the caregiver

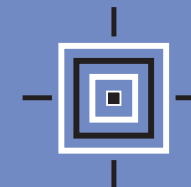
- » Other treatments could be simple remedies that can be followed together with the drugs to help recovery.
- » The health worker should discuss and help the caregiver understand these other treatments.
- » The health worker should counsel the caregiver in terms of feeding the child and when to return urgently.

3. Follow-up

- » The system will suggest when the caregiver should follow up the child.
- » The health worker should share these details with the caregiver and explain when to return immediately.

4.3 Additional considerations for adapting workflows

These workflows are meant to be generic and high-level. They require a level of customization and adaptation when they are translated into a digital system for a specific context. These workflows are 80% complete; the other 20% will need to be done through a series of human-centred design and mechanisms to complete the workflows for implementation. For example, additional workflows may be needed or there might be additional activities expected of a health worker in the facility. Some workflows are not included because of the high level of contextualization required, dispensing (if separate from service provision), configuration (of facility-level specifics) and follow-up (which can be automated). Alternatively, there might be some activities and tasks that a health worker would not be expected to do. Although these workflows can be considered as a starting point, it is helpful to conduct further validation through interviews with the targeted personas or shadowing their work to obtain a better sense of the differences that should be reflected in the digital system.



This section outlines the minimum set of data corresponding to different points of the workflow within the identified business processes. This data set can be used with any software system and lists the data elements relevant for service delivery and for executing the decision-support logic, as well as populating indicators and performance metrics. Although this section provides a high-level overview of the data elements, a more complete data dictionary in spreadsheet form detailing the input options, validation checks and concept dictionary codes is available [here](#).

Inclusion of a data element in the table does not by itself indicate that data collection is required. Additionally, some data elements are dependent on other data elements (e.g. test results are only entered when a test has been performed). Collection of data should not prevent clients from receiving treatment or affect clinical care.

5.1 Simplified list of core data elements

Table 9 provides a simplified list of core data elements and is merely a snapshot of the comprehensive data dictionary found [here](#). As with the workflows, this data dictionary is 80% generic with the expectation that the other 20% will be supplemented and modified via country adaptation.

Table 9: Workflow core data elements for the identified business processes

Activity ID. Activity name	Data element ID	Data element label	Description and definition/user information
Business process CHE.A. Registration			
CHE.A1. Arrive at the facility	N/A – No data are recorded during this activity		
CHE.A4. Gather the child's details	CHE.A.DE01	National unique identification	Unique identifier provided or a universal ID, if used in the country
	CHE.A.DE02	First name	Child's first name
	CHE.A.DE03	Middle name	Child's middle name
	CHE.A.DE04	Last name	Child's family name or last name
	CHE.A.DE05	Date of birth	The child's date of birth, if known
	CHE.A.DE06	Time of birth	The child's time of birth (if <1 day old)
	CHE.A.DE07	Age	Age of the child auto-calculated (presented as the number of years, months, days) based on their date of birth
	CHE.A.DE16	Sex	Sex of the child
	CHE.A.DE20	Caregiver first name	The first name of the child's caregiver
	CHE.A.DE21	Caregiver middle name	The middle name of the child's caregiver
	CHE.A.DE22	Caregiver last name	The last name of the child's caregiver
	CHE.A.DE30	Caregiver's mobile number	The mobile phone number of the child's caregiver
	CHE.A.DE34	Biological mother's vital status	The vital status of the child's mother
CHE.A3. Search for the patient	N/A – No data are recorded during this activity		
CHE.A5. Create the child's record	N/A – No data are recorded during this activity; see CHE.A4. Gather the child's details		
CHE.A6. Validate the client's details	N/A – No data are recorded during this activity		
CHE.A7. Check in the client	N/A – No data are recorded during this activity		

Activity ID. Activity name	Data element ID	Data element label	Description and definition/user information
Business process CHE.B. Assess and classify the sick child			
CHE.B3. Encounter	CHE.B3.DE01	Determine reason for the consultation	The reason for the child's consultation.
	CHE.B3.DE06	Type of consultation	The child's visit is for a new consultation.
CHE.B7. Danger signs	CHE.B7.DE01	Obstructed or absent breathing	The child has obstructed or absent breathing. Treat immediately .
	CHE.B7.DE02	Convulsing now	The child is convulsing now. Treat immediately.
	CHE.B7.DE04	Unconscious or lethargic	A lethargic child is not awake and alert when they should be. The child is drowsy and does not show interest in what is happening around them. Often the lethargic child does not look at their caregiver or watch your face when you talk, or will not respond if you clap or snap your fingers. The child may stare blankly and appear not to notice what is going on around them. An unconscious child cannot be wakened. They do not respond when they are touched, shaken or spoken to. Ask the caregiver if the child seems unusually sleepy or if they cannot wake the child. Look to see if the child awakens when the caregiver talks to or shakes the child or when you clap your hands.
	CHE.B7.DE05	Not able to drink or breastfeed	A child shows signs of not being able to drink or breastfeed if they are not able to suck or swallow when offered a drink or breast-milk.
CHE.B6. Measurements	CHE.B6.DE01	Axillary temperature (°C)	The child's axillary temperature (temperature taken under the armpit), measured in °C.
	CHE.B6.DE02	Health worker prefers taking the rectal temperature	The health worker prefers to take the child's rectal temperature.
	CHE.B6.DE03	Rectal temperature (°C)	The child's rectal temperature, measured in °C.
	CHE.B6.DE04	Measured temperature	Auto-calculated temperature range based on the axillary temperature in °C (add 0.5 for rectal temperature).
	CHE.B6.DE05	Thermometer not available	A thermometer is not available to measure the child's temperature.
	CHE.B6.DE06	Hot to the touch	The child is hot to the touch.
	CHE.B6.DE07	Weight (kg)	The child's weight in kg, to one decimal place.
CHE.B10.Cough or difficulty breathing	CHE.B10S1.DE05A	Cough	The caregiver reports that the child has a cough, or you see that the child has a cough.
	CHE.B10S1.DE01	Difficulty breathing	Difficult breathing is any unusual pattern of breathing. Caregivers describe this in different ways. They may say that their child's breathing is fast or noisy or interrupted. If a mother answers no, look to see if you think the child has cough or difficult breathing.
	CHE.B11S1.DE01	Diarrhoea	In children, diarrhoea is defined as three or more loose or watery stools in a 24-hour period. In young infants, diarrhoea is defined as stools that have changed from the usual pattern and are many and watery. This means more water than faecal matter. The normally frequent or semi-solid stools of a breastfed baby are not diarrhoea.

N/A: not applicable.

5.2 List of calculated data elements

The previous section outlined the core data elements that should be included within digital systems to facilitate the decision-support logic or indicators. There are additional data-derived elements that are based on calculations from the core data elements.

Table 10: Calculated data elements

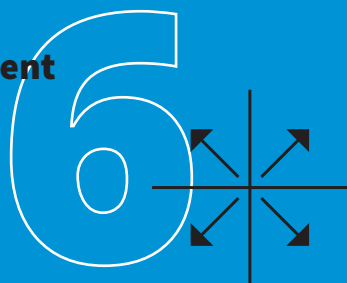
Calculated data element label	Core data elements used for the calculation (i.e. the variables)	Calculation
Fast breathing	» Respiratory rate second count » Respiratory rate	IF (“Age” < 2 ‘months’ and (“Respiratory rate second count” ≥ 60 ‘breaths per minute’) or (“Respiratory rate first count” ≥ 60 ‘breaths per minute’ and “Respiratory rate second count not possible” = true) OR (“Age” ≥ 2 ‘months’ and “Age” < 12 ‘months’ and “Respiratory rate” ≥ 50 ‘breaths per minute’) OR (“Age” ≥ 12 ‘months’ and “Age” < 60 ‘months’ and “Respiratory rate” ≥ 40 ‘breaths per minute’)
Age	» Date of birth » Visit date	(‘Visit date’ – ‘Date of Birth’)/365.25
Measured temperature	» Axillary temperature » Rectal temperature	Auto-calculated based on axillary temperature Low = IF “Axillary temperature” < 35.5°C Normal = IF (“Age” < 2 ‘months’ AND “Axillary temperature” ≥ 35.5°C AND “Axillary temperature” < 38.0°C) OR (“Age” ≥ 2 ‘months’ AND “Axillary temperature” ≥ 35.5°C AND “Axillary temperature” < 37.5°C) High = IF (“Age” < 2 ‘months’ AND “Axillary temperature” ≥ 38.0°C) OR (“Age” ≥ 2 ‘months’ AND “Axillary temperature” ≥ 37.5°C AND “Axillary temperature” < 38.5°C) Very high = IF “Age” ≥ 2 ‘months’ AND “Axillary temperature” ≥ 38.5°C Based on rectal temperature: Low = IF “Rectal temperature” < 36°C Normal = IF (“Age” < 2 ‘months’ AND “Rectal temperature” ≥ 36°C AND “Rectal temperature” < 38.5°C) OR (“Age” ≥ 2 ‘months’ AND “Rectal temperature” ≥ 36°C AND “Rectal temperature” < 38°C) High = IF (“Age” < 2 ‘months’ AND “Rectal temperature” ≥ 38.5°C) OR (“Age” ≥ 2 ‘months’ AND “Rectal temperature” ≥ 38°C AND “Rectal temperature” < 39°C) Very high = IF “Age” ≥ 2 ‘months’ AND “Rectal temperature” ≥ 39°C

5.3 Additional considerations for adapting the data dictionary

Some settings may require the inclusion of additional data elements into the full data set or changes to the response options based on contextual differences. Additionally, the transition from paper-based forms to digital systems may require some reflection on whether data elements currently on the paper forms should be incorporated into the digital system. [Table 11](#) is an initial list of considerations anticipated for each implementation to review and customize based on the national guidelines and local context.

Table 11: Characteristics for local customization and configuration

Points of customization and configuration	Description
Population mean	The country's population mean
HIV prevalence	The prevalence of HIV in the specific country
Malaria risk	The risk of malaria in the specific country
Unique identifier	This can be modified according to the country format or accepted documents for ID
Facility identifier	Unique identifier of the facility. A reference to a facility registry or a reporting system (e.g. DHIS2) should be included where possible
Facility name	Name of the different health-care facilities based on a facility registry or a reporting system (e.g. DHIS2) should be included where possible



Decision-support logic

The decision-support logic component of the DAK provides the decision logics and algorithms, and the scheduling of services, in accordance with WHO guidelines. In this DAK, the decision logic and algorithms deconstruct the recommendations within the child health in humanitarian emergencies guidelines and guidance into a format that clearly labels the inputs and outputs that would be operationalized in a digital decision-support system.

6.1 Decision-support logic overview

Table 12 provides an overview of the decision-support tables and algorithms for the different child health in humanitarian emergencies business processes. The structure of the decision-support tables is based on an adaptation of the Decision Model and Notation, an industry standard for modelling and executing decision logics (26). These decision-support tables detail the business rules, data inputs and outputs to support child health in humanitarian emergencies module business processes.

Table 12: Overview of decision-support tables for child health in humanitarian emergencies module

Activity ID. Activity name	Decision-support table ID	Decision-support table description	Reference or source
CHE.B. Assess and classify the sick child	CHE.DT.01	Classifications: ≥ 2 months to 59 months	<i>Integrated management of childhood illness (chart booklet) (March 2014)</i> (https://iris.who.int/handle/10665/104772) (15)
CHE.B. Assess and classify the sick child	CHE.DT.02	Classifications: young infant	<i>Management of the sick young infant aged up to 2 months: IMCI chart booklet</i> (https://iris.who.int/handle/10665/326448) (16)
CHE.C. Treat the sick child	CHE.DT.03	Treatment: ≥ 2 months to 59 months	<i>Integrated management of childhood illness (chart booklet) (March 2014)</i> (https://iris.who.int/handle/10665/104772) (15)

6.2 Decision-support tables

Each of the decision logics listed in the overview table is elaborated [here](#). These decision-support tables comprise the components described in [Table 13](#).

Note that the decision-support logic has been translated directly from WHO guidelines and guidance documents and has been reviewed by the panel of experts who created those guidelines documents. The decision-support logic has been created and reviewed by clinical experts. However, we anticipate this to be further localized at the country level; some level of adaptation may be needed depending on changes to the workflow or changes to the data dictionary.

Any changes to the decision-support logic should be considered carefully because an embedded decision-support system can greatly affect the quality of care at the point of care. As helpful as a decisions-support logic can be to the health worker, an incorrect decision-support logic can also be detrimental. Thus, any new decision-support logic should be carefully reviewed and agreed on by in-country clinical experts.

Table 13: Components of the decision-support tables

Decision ID		The name of the decision describing which algorithm or logic is represented (e.g. reaching a classification of severe dehydration for older children). The decision ID should correspond to the number in Table 7.		
Business rule		The description of the decision that needs to be made based on IF/THEN statements with the appropriate data element label for your variables. The rule should demonstrate the relationship between the input variables and the expected outputs and actions within the decision logic.		
Trigger		The event that would indicate when this decision-support logic should appear within the workflow, such as the activity that would trigger this decision to be made.		
Input(s)		Output	Action	Annotations
These are the variables that need to be considered to determine the consequent actions or outputs.	If there are multiple input entries in the same row, these different inputs are considered as “AND” conditions that need to be in place at the same time.	The resulting action or decision based on the combination of input entries. This is the statement that immediately follows “THEN”. Examples of outputs may include a diagnosis, alerts or prompts for referral or counselling.	Concrete measures to be taken based on the output (e.g. refer, provide treatment or counselling, conduct test). In some cases, the output and action may be the same. This is where you would write the content that would appear in your pop-up message, which would notify the health worker of the appropriate next steps; these can include counselling, a case management approach or referral.	Additional explanations or descriptions, including possible pop-up alert messages and relevant background information. This section can also include the written content (in addition to the action) that would appear in your pop-up messages that would notify the health worker of the appropriate next steps, which can include counselling, a case management approach or referrals.
	Inputs placed in different rows are considered as OR and can be considered independently of the inputs in other rows.			

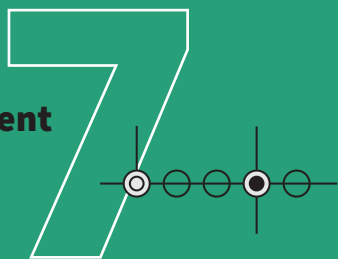
Table 14 applies the decision-support logic to the classification of older children.

Table 14: Example decision-support logic table for the classification of older children

Decision ID		CHE.DT.01				
Business rule		Classify ≥2 months and <59 months				
Trigger		‘Signs’ for age ≥2 months to <60 months completed				
Inputs			Outputs	Action	Annotations	References
“Danger signs” = FALSE	“Obstructed or absent breathing” = TRUE	“Continue to assess sick child” = “Stabilized, continue consultation”	Identify “Classification” = “CHE.B23.DE02 – History of obstructed or absent breathing”	Complete “Treatment” for age ≥2 months to <60 months	Health worker to use the proposed treatment to treat the child. In case of severe classifications, the child is to be referred to a nearby hospital	<i>Integrated management of childhood illness (chart booklet) (March 2014) (15)</i> (https://iris.who.int/handle/10665/43993)
“Convulsing now” = TRUE	“Continue to assess sick child” = “Stabilized, continue consultation”		Identify “Classification” = “CHE.B23.DE01 – very severe disease”	Complete “Treatment” for age ≥2 months to <60 months		
“Danger signs” = TRUE	“Cough” = TRUE		Identify “Classification” = “CHE.B23.DE03 – severe pneumonia or very severe disease”	Complete “Treatment” for age ≥2 months to <60 months		
“Diarrhoea” = TRUE	“Diarrhoea for how long?” = “14 days or more”	“Classification” = “No dehydration”	Identify “Classification” = “CHE.B23.DE17 – persistent diarrhoea”	Complete “Treatment” for Age ≥2 months to <60 months	Health worker to use the proposed treatment to treat the child. Tell the caregiver about the medication, other treatment and advise on the required follow-up.	

For all the decision-support tables that are available for the child health in humanitarian emergencies DAK, please refer to [this link](#).

Component



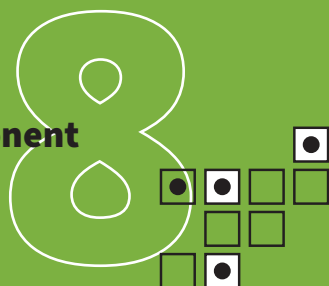
Indicators and performance metrics

This section outlines the indicators and performance metrics that would be aggregated from the core data elements identified in [Component 5](#). The list in [Table 15](#) is a minimum set of indicators that can be aggregated for decision-making, performance metrics and subnational and national reporting based on data collected from individual-level, routine health systems. These indicators may be aggregated automatically from the digital tracking tool to populate a digital HMIS, such as the DHIS2. This table is also available [here](#).

Table 15: Indicators and performance metrics

Indicator ID	Indicator name	Definition	Numerator definition	Denominator description	Disaggregation
CHE.IND.1	Severe illness, all children	Proportion of consultations with infants and children up to 5 years old with severe disease/severe illness	Number of consultations with infants and children aged 0–59 months meeting the criteria for any severe illness/severe disease classification	Number of consultations with infants and children aged 0–59 months	Age (<2months, 2–59 months), sex, facility, area/region, time period
CHE.IND.2	Possible serious bacterial infection (PSBI), young infants	Proportion of consultations with young infants under 2 months old with signs of PSBI or very severe disease	Number of consultations with infants under 2 months of age who meet the criteria for PSBI classification	Number of consultations with infants under 2 months of age	Age (<7 days, 7–27 days, 28 to <59 days), facility, sex, area/region, time period
CHE.IND.3	Danger signs, children aged 2–59 months	Proportion of consultations with children aged 2 months and up to 5 years with danger signs	Number of consultations with infants and children aged 2 months to 5 years with one or more danger signs	Number consultations with children aged 2–59 months	Sex, facility, area/region, time period
CHE.IND.4	Pneumonia, all children	Proportion of consultations with infants and children aged up to 5 years with pneumonia	Number of consultations with infants and children aged 0–59 months who meet the criteria for pneumonia classification	Number of consultations with infants and children aged 0–59 months	Age (<2months, 2–59 months), sex, facility, area/region, time period
CHE.IND.5	Pneumonia, children with a cough/difficulty breathing	Proportion of consultations with children aged 2 months up to 5 years with a cough or difficulty breathing with pneumonia	Number of consultations of infants and children aged 2–59 months who meet the criteria for pneumonia classification	Number of consultations with children aged 2–59 months with cough or difficulty breathing	Age (<2months, 2–59 months), sex, facility, area/region, time period

Component



High-level functional and non-functional requirements

This section provides an overview of illustrative functional and non-functional requirements that may be considered to kick-start the process of designing or adapting the DTDS system. Functional requirements describe the capabilities the system must have to meet the needs of the end users and achieve the tasks within the business process. Non-functional requirements provide the general attributes and features of the digital system to ensure usability and overcome technical and physical constraints. Examples of non-functional requirements include ability to work offline, multiple language settings and password protection.

Table 16 highlights some key functional requirements for executing the business processes listed in **Component 4** of this DAK; the complete set of functional requirements can be accessed [here](#). **Table 17** provides non-functional requirements as general characteristics of the overall system. Please note that these are not exhaustive lists and should be modified according to context and user persona needs.

8.1 Functional requirements

Table 16: Key functional requirements

Requirement ID	Activity ID and description	As a ...	I want to ...	So that ...
Business process A: Registration				
CHE.FXREQ.001	Infant is born	Neonate/infant (caregiver)	N/A	N/A
CHE.FXREQ.002	Sick neonate/infant arrives at health facility	Neonate/infant (caregiver)	N/A	N/A
CHE.FXREQ.003	Sick neonate/infant is born	Neonate/infant (caregiver)	N/A	N/A
CHE.FXREQ.004	Sick child arrives at facility	Child (caregiver)	N/A	N/A
CHE.FXREQ.005	Triage and emergency assessment	Health worker or clerk	by-pass the standard flow at any point if an urgent case is identified. Refer to business process G. Triage and assessment	the child/infant can receive immediate attention. Refer to business process G. Triage and assessment
CHE.FXREQ.006	Gather neonate/infant/child details	Health worker or clerk	be able to gather basic information about the child's caregiver (if required)	I can track and follow the name of the client's caregiver for the purposes of engagement during the consultation

Requirement ID	Activity ID and description	As a ...	I want to ...	So that ...
CHE.FXREQ.007	Gather neonate/infant/child details	Health worker or clerk	be able to gather unique identifier number/ information pertaining to the client that does not include personal information	I do not have to deal with personal information management regulations/laws
CHE.FXREQ.008	Gather neonate/infant/child details	Health worker or clerk	be able to gather unique identifier number/ information pertaining to the client that includes personal information	I can identify the client using personal information within personal information management regulations/laws
CHE.FXREQ.009	Search for neonate/infant/child record	Health worker or clerk	search if the client is already in the system (using at least two identifiers) (including information pertaining to the child's caregiver)	I can check if this is a new or existing client
CHE.FXREQ.010	Search for neonate/infant/child record	Health worker or clerk	display sufficient data to identify the client	I can confirm that it is the correct client
CHE.FXREQ.011	Search for neonate/infant/child record	Health worker or clerk	search if a client is already in the system before starting a new medical record entry	I can confirm that it is the correct client and update information as necessary
CHE.FXREQ.012	Search for neonate/infant/child record	Health worker or clerk	read client identification system (e.g. quick response [QR] code, barcode, fingerprint) and pull up the information about the client	I can confirm that it is the correct client and update information as necessary
CHE.FXREQ.013	Search for neonate/infant/child record	Health worker or clerk	provide sufficient data to rule out that this client is already in the system	I can avoid duplicates

N/A: not applicable.

8.2 Non-functional requirements

Table 17: Non-functional requirements

Requirement ID	Category	Non-functional requirement
CHE.NFXNREQ.001	Security – confidentiality	Provide password-protected access for authorized users
CHE.NFXNREQ.002	Security – confidentiality	Provide a means to ensure confidentiality and privacy of personal health information
CHE.NFXNREQ.003	Security – confidentiality	Provide ability for allowed users to view confidential data
CHE.NFXNREQ.004	Security – confidentiality	Anonymize data that are exported from the system when required (also ability to export non-anonymized data from the system)
CHE.NFXNREQ.005	Security – confidentiality	Prevent remembering username and password
CHE.NFXNREQ.006	Security – confidentiality	Automatically log out the user after a specified time of inactivity
CHE.NFXNREQ.007	Security – confidentiality	Provide encrypted communication between components
CHE.NFXNREQ.008	Security – authentication	Notify the user to change their password the first time they log in
CHE.NFXNREQ.009	Security – authentication	Adhere to complex password requirements
CHE.NFXNREQ.010	Security – authentication	Provide a mechanism to securely change a user's password

Requirement ID	Category	Non-functional requirement
CHE.NFXNREQ.011	Security – authentication	Notify the user of a password change to their account
CHE.NFXNREQ.012	Security – authentication	Reset a user's password in a secure manner
CHE.NFXNREQ.013	Security – authentication	Lock a user out after a specified number of wrong password attempts
CHE.NFXNREQ.014	Security – authentication	Notify a user if their account is locked due to wrong password attempts
CHE.NFXNREQ.015	Security – authentication	Provide role-based access to the system
CHE.NFXNREQ.016	Security – audit trail and logs	Log system log-ins and log-outs
CHE.NFXNREQ.017	Security – audit trail and logs	Record all authentication violations
CHE.NFXNREQ.018	Security – audit trail and logs	Log all activities performed by the user, including date and time stamp
CHE.NFXNREQ.019	Security – audit trail and logs	Log access to views of individual client records
CHE.NFXNREQ.020	Security – audit trail and logs	Log access to data summaries, reports, analysis and visualization features
CHE.NFXNREQ.021	Security – audit trail and logs	Log exchange of data with other systems
CHE.NFXNREQ.022	Security – audit trail and logs	Generate analysis of the use of different system features and reports
CHE.NFXNREQ.023	Security – audit trail and logs	Log all data and system errors
CHE.NFXNREQ.024	Security – user management	Allow user with permission to create a new user and temporary password
CHE.NFXNREQ.025	Security – user management	Provide role-based access
CHE.NFXNREQ.026	Security – user management	Allow roles to be associated with specific geographical areas or health-care facilities (with the ability to choose which facility to log into as the user may work across several facilities)
CHE.NFXNREQ.027	Security – user management	Allow cascading user management and assignment of roles
CHE.NFXNREQ.028	Security – user management	Allow user to change the user's own password
CHE.NFXNREQ.029	Security – user management	Allow users to enable and disable another user
CHE.NFXNREQ.030	Security – user management	Allow admin user to request password reset
CHE.NFXNREQ.031	Security – user management	Notify the user to regularly change the user's password
CHE.NFXNREQ.032	Security – user management	Allow each user to be assigned to one or more roles
CHE.NFXNREQ.033	Security – user management	Support definitions of unlimited roles and assigned levels of access, viewing, entry, editing and auditing
CHE.NFXNREQ.034	System requirements – general	Provide a unique version number for each revision
CHE.NFXNREQ.035	System requirements – general	Enable earlier versions of a record to be recoverable
CHE.NFXNREQ.036	System requirements – general	Enable deployment in an environment subject to power loss
CHE.NFXNREQ.037	System requirements – general	Work in an environment that is subject to loss of connectivity
CHE.NFXNREQ.038	System requirements – general	Generate IDs that are unique across different installations or sites
CHE.NFXNREQ.039	System requirements – general	Report version number when saving data to the database

Requirement ID	Category	Non-functional requirement
CHE.NFXNREQ.040	System requirements – general	Be designed to be flexible enough to accommodate necessary changes in the future
CHE.NFXNREQ.041	System requirements – general	Allow for offline and online functionality
CHE.NFXNREQ.042	System requirements – general	Show the number of records that are not yet synchronized
CHE.NFXNREQ.043	System requirements – general	Have the ability to easily back up information
CHE.NFXNREQ.044	System requirements – general	Warn user if no valid backup for more than a predefined number of days
CHE.NFXNREQ.045	System requirements – general	Must have the ability to store images and other unstructured data
CHE.NFXNREQ.046	System requirements – scalability	Scalable to accommodate new demands
CHE.NFXNREQ.047	System requirements – scalability	Be able to accommodate at least 3000 health-care facilities
CHE.NFXNREQ.048	System requirements – scalability	Be able to accommodate at least 5000 concurrent users
CHE.NFXNREQ.049	System requirements – scalability	Be able to accommodate more than [x number of] concurrent users
CHE.NFXNREQ.050	System requirements – usability	Be user-friendly for people with low computer literacy
CHE.NFXNREQ.051	System requirements – usability	Provide informative error messages and tool tips
CHE.NFXNREQ.052	System requirements – usability	Alert the user when navigating away from the form without saving
CHE.NFXNREQ.053	System requirements – usability	Support real-time data entry validation and feedback to prevent data entry errors from being recorded
CHE.NFXNREQ.054	System requirements – usability	Simplify data recording through predefined dropdown or searchable lists, radio buttons and check boxes
CHE.NFXNREQ.055	System requirements – usability	Support multiple languages – English, French, Spanish, Chinese, Russian, Arabic
CHE.NFXNREQ.056	System requirements – usability	Use industry-standard user interface practices and apply them in a consistent manner throughout the system
CHE.NFXNREQ.057	System requirements – usability	Easy to learn and intuitive to enable the user to navigate between pages
CHE.NFXNREQ.058	System requirements – usability	Provide guidance to the users to better support clinical guidelines and best clinical practices

Requirement ID	Category	Non-functional requirement
CHE.NFXNREQ.059	System requirements – usability	Be reliable and robust (minimize the number of system crashes)
CHE.NFXNREQ.060	System requirements – usability	Adjust display to fit small screens (such as mobile phones)
CHE.NFXNREQ.061	System requirements – configuration	Configure the system centrally
CHE.NFXNREQ.062	System requirements – configuration	Configure business rules in line with guidelines and standard operating procedures (using the decision logic and contextual query language (CQL) + Fast Health Interoperability Resources [FHIR])
CHE.NFXNREQ.063	System requirements – configuration	Configure error messages
CHE.NFXNREQ.064	System requirements – configuration	Configure workflows and business rules to accommodate differences between facilities
CHE.NFXNREQ.065	System requirements – interoperability	Communicate with external systems through mediators
CHE.NFXNREQ.066	System requirements – interoperability	Provide access to data through application programming interfaces to interface with other SMART guideline reference app solutions, incumbent solutions (e.g. registration systems) and reporting tools
CHE.NFXNREQ.067	System requirements – interoperability	The reference application must have an FHIR back end/FHIR native and leverage the workstream 3 content (CQL + FHIR resources)
CHE.NFXNREQ.068	System requirements – interoperability	Be interoperable with external systems through mediators
CHE.NFXNREQ.069	System requirements – interoperability	Link with insurance systems to verify eligibility and submit claims
CHE.NFXNREQ.070	System requirements – interoperability	Exchange data with other approved systems
CHE.NFXNREQ.071	System requirements – interoperability	Accept data from multiple input methods including paper and geocoding (global positioning system)
CHE.NFXNREQ.072	System requirements – hardware and connectivity	Allow for data exchange and efficient synchronization across multiple facilities and points of service when the internet is available, even when it is intermittent and slow
CHE.NFXNREQ.073	System requirements – hardware and connectivity	Enable use via a web interface and mobile applications for use on various technology devices such as computers, laptops, tablets and mobiles
CHE.NFXNREQ.074	System requirements – hardware and connectivity	Ability to work offline and in low-bandwidth areas
CHE.NFXNREQ.075	Documentation	Provision of design, build, technical system and architecture, and testing documentation, for future support and further configuration of the solution for other health intervention initiatives

References

- Digital implementation investment guide (DIIG): integrating digital interventions into health programmes. Geneva: World Health Organization; 2020 (<https://iris.who.int/handle/10665/334306>).
- Digital health. In: Seventy-first World Health Assembly WHA71.7, 26 May 2018 (<https://iris.who.int/handle/10665/279505>).
- Global strategy on digital health 2020–2025. Geneva: World Health Organization; 2021 (<https://iris.who.int/handle/10665/344249>).
- World Bank Group, United States Agency for International Development, World Health Organization. The roadmap for health measurement and accountability (MA4Health): a common agenda for the post-2015 era. Washington (DC): World Bank Group; 2015 (<https://www.healthdatacollaborative.org/fileadmin/uploads/hdc/Documents/the-roadmap-for-health-measurement-and-accountability.pdf>).
- United States Agency for International Development. Digital strategy 2020–2024. Washington (DC): United States Agency for International Development; undated (https://www.edu-links.org/sites/default/files/media/file/USAID_Digital_Strategy.pdf).
- The Global Fund. The Global Fund strategic framework for data use for action and improvement at country level, 2017–2022. Geneva: The Global Fund to Fight AIDS, Tuberculosis and Malaria (https://www.theglobalfund.org/media/8362/me_datauseforactionandimprovement_framework_en.pdf).
- Recommendations on digital interventions for health system strengthening: evidence and recommendations. Geneva: World Health Organization; 2019 (<https://iris.who.int/handle/10665/311980>).
- Classification of digital health interventions v1.0: a shared language to describe the uses of digital technology for health. Geneva: World Health Organization; 2018 (<https://iris.who.int/handle/10665/260480>).
- Barton C, Kallem C, Van Dyke P, Mon D, Richesson R. Demonstrating “collect once, use many” – assimilating public health secondary data use requirements into an existing Domain Analysis Model. AMIA Annu Symp Proc. 2011;2011:98–107.
- Mehl G, Tunçalp Ö, Ratanaprayul N, Tamrat T, Barreix M, Lowrance D, et al. WHO SMART guidelines: optimising country-level use of guideline recommendations in the digital age. Lancet Digit Health. 2021;3:e213–6. doi: 10.1016/S2589-7500(21)00038-8.
- Boxwala AA, Rocha BH, Maviglia S, Kashyap V, Meltzer S, Kim J, et al. A multi-layered framework for disseminating knowledge for computer-based decision support. J Am Med Inform Assoc. 2011;18(Suppl 1):i132–9. doi:10.1136/amiajnl-2011-000334.
- UHC compendium: repository of interventions for universal health coverage [website]. Geneva: World Health Organization; 2020 (<https://www.who.int/universal-health-coverage/compendium/interventions-by-programme-area>).
- National eHealth strategy toolkit. Geneva: World Health Organization and International Telecommunication Union; 2012 (<https://iris.who.int/handle/10665/75211>, accessed 3 November 2023).
- WHO recommendations: optimizing health worker roles to improve access to key maternal and newborn health interventions through task shifting (OptimizeMNH). Geneva: World Health Organization; 2012 (<https://iris.who.int/handle/10665/77764>).
- Integrated management of childhood illness (chart booklet) (March 2014). Geneva: World Health Organization; 2008 (<https://iris.who.int/handle/10665/43993>).
- Management of the sick young infant aged up to 2 months: participant manual. Geneva: World Health Organization; 2019 (<https://iris.who.int/handle/10665/326644>).
- Integrated management of childhood illness: management of the sick young infant aged up to 2 months: IMCI chart booklet. Geneva: World Health Organization; 2019 (<https://iris.who.int/handle/10665/326448>).
- Integrated management of childhood illness: distance learning course. Geneva: World Health Organization; 2014 (<https://iris.who.int/handle/10665/104772>).
- Emergency triage assessment and treatment (ETAT). Geneva: World Health Organization; 2005 (<https://iris.who.int/handle/10665/43386>).
- Guideline on the prevention and management of wasting and nutritional oedema (acute malnutrition). Geneva: World Health Organization; 2023 (<https://app.magicapp.org/#/guideline/7352>).
- Pocket book of hospital care for children: guidelines for the management of common childhood illnesses, 2nd ed. Geneva: World Organization; 2013 (<https://iris.who.int/handle/10665/81170>).
- Integrated management of childhood illness: caring for newborns and children in the community. Geneva: World Health Organization; 2011 (<https://iris.who.int/handle/10665/44398>).
- Classifying health workers: mapping occupations to the international standard classification. Geneva: World Health Organization; 2019.
- International Labour Organization (ILO). The International Standard Classification of Occupations (ISCO). Geneva: ILO; 2016 (<https://www.ilo.org/public/english/bureau/stat/isco/isco08/index.htm>, accessed 19 August 2023).
- HL7 FHIR Release 5 [website]. 2.24.0 Data types. Ann Arbor (MI): Health Level 7 International; 2023 (<http://hl7.org/fhir/datatypes.html>).
- Object Management Group (OMG). Decision Model and Notation version 1.5 beta. Milford (CT): OMG; 2020 (<https://www.omg.org/spec/DMN>, accessed 19 August 2023).
- Developing health management information systems: a practical guide for developing countries. Manila: World Health Organization Regional Office for the Western Pacific; 2004 (<https://iris.who.int/handle/10665/207050>).
- Public Health Informatics Institute. Collaborative Requirements Development Methodology (CRDM). In: Public Health Informatics Institute [website]. Decatur (GA): The Task Force for Global Health; 2016 (<https://www.phii.org/crdm/>, accessed 19 August 2023).

Annex. Guidance for adding data elements to or amending existing data elements in the data dictionary

When adapting the data dictionary, data elements may need to be modified or added because of the structure of existing paper registers or local reporting requirements. If starting from paper-based registers and forms, you can find additional guidance in the *WHO recommendations: optimizing health worker roles to improve access to key maternal and newborn health interventions through task shifting (OptimizeMNH) (1)*. Below is an overview of the data mapping to provide a template for standardizing the data dictionary. When amending the wording of data elements, it is important to ensure that the standard terminology codes still reflect the data element as originally intended.

What to note	Description
Activity ID	The ID number of the task in the workflow in which the data element is collected. This will denote the point in time at which this data element is collected. This should align with the Activity ID that is provided in form mapping.
Data element code	Each data element should have an identification number or code that is unique across the entire project. Use existing serial or identification numbers when available. If no identifiers exist, fields should be enumerated in a logical format.
Form ID and form data element label	The Form ID should be from the ID listed. List the Form ID in which the data element appears. This is important for ensuring that the design of the digital system has taken into account all the required paper forms and data elements on those paper forms. Also list the label of the data element as written on the original form (or translated as closely as possible). This will be key in keeping track of which data elements from the original paper forms are duplicated. Note that duplicate data fields can be included purposely. (Patient identifiers, such as name, date of birth, village and ID number, would be included in multiple data instruments to identify an individual patient.)
Data element label	The label of the data element written in a way in which the end users can easily understand, for example, education level, weight, height, reasons for coming into facility, which medication(s) your client is taking. The data element label in this column is what will be used in the digital form. The digital register should not simply replace the paper registers, but it should also streamline processes and link duplicated data elements; thus the data element label listed here should be what will be used in the digital system.
Data element name	The shorthand name for the data element (e.g. educ_level for education level; weight_kg for weight; height_m for height). This will be key when coding the system and determining the calculations required. This data element name is what will reconcile any duplicate data elements in the digital system.

What to note	Description
Description and definition	The description of the data field, including any units that define the field (e.g. weight in kilograms [kg]). Provide a clear explanation of what this data point is requesting, assuming the reader has never seen the form. Be sure to use consistent and easy-to-understand terminology across all forms. This is particularly important if the data element name differs across forms but requires the same input.
Data type	The data type should be aligned with data types outlined by the Fast Health Interoperability Resources (FHIR) standards. Some common data types are: » Boolean (e.g. TRUE/FALSE, YES/NO); » string (a sequence of Unicode characters, e.g. name); » date (e.g. date of birth); » time (e.g. time of appointment); » ID (e.g. unique identifier assigned to the client); » integer (a whole number, e.g. the number of previous appointments); » decimal (rational numbers that have a decimal representation, e.g. exact duration of time, location coordinates, all percentages); » observation (health status observations collected from the client, e.g. height, weight, eye colour, pulse, blood pressure, temperature, glucose level); » codable concept (a value that is usually supplied by providing a reference to one or more data points, e.g. body mass index [BMI], contraceptive prevalence rate); » signature (an electronic representation of a signature that is either cryptographic or a graphical image that represents a signature or a signature process, e.g. supervisor's approval); » attachment (additional data content defined in other formats, e.g. images); » note that if there are multiple choice data fields, the parent data field should be labelled "Multiple choice – Select one" or "Multiple choice – Select all that apply". Then each individual option should be listed in the input option column and be classified with one of the data types listed above. Although this list should be sufficient to relay this information to a health informatics specialist or technology vendor, there are many more data codes that can be applied to achieve a more precise classification. For other possible data types, please refer to the HL7 FHIR guide on data types (2).

What to note	Description
Input options	For multiple choice fields only; otherwise leave this column blank. Write the list of responses from which the health worker may select. Each of these input options should be in a separate row and reference the parent data ID in its own data ID (e.g. [parent data element ID].[option number]). Each of these options should be labelled with a data type as indicated above.
Calculation	For codable concept data type fields, write the formula that defines the field. Leave this column blank for all other data types. Use standard mathematical symbols and the data element label of the data element names included in the formula (e.g. for the BMI calculation: weight_kg/[height_m]2).
Validation required	Yes or No, to indicate whether there needs to be some form of validation. The constraints provided by a range of acceptable responses are mentioned in next column – Validation condition
Validation condition	The range of acceptable responses, if validation is required (e.g. for a phone number, only 10 digits are allowed; for a birthday, only past dates are allowed).
Editable	Yes or No to indicate whether the end user, or health worker, would be able to edit the field after it has been entered into the system.
Required	Note whether or not this field is: » Required – R » Required but can be left empty – RE » Optional – O » Conditional on answers from other data fields – C » Conditional but can be left empty – CE
Reason for the required data	If this field is required (R), state the reason whether for: » accountability for national-level reporting; » service delivery or clinical decision-making; » client identification. The digital system should not simply replace your paper registers but it should also streamline processes; thus, it is important to understand why a certain data field is actually required and seek opportunities to optimize data flows. Given the high volume of data collection required of health workers, it might be better to remove a data entry field if it serves no real purpose for the clinician, public health reporting or any other identified purpose.

What to note	Description
Skip logic	If this field is conditional on answers from other data fields (C) or conditional but can be left empty (CE), denote what the skip logic is. This is common for data elements that are a part of follow-up questions. If the input of one data element field has a value lower than a certain threshold, then some data inputs can be skipped. Those data elements will have skip logic that is defined by a preset threshold.
Linked to aggregate indicator	Aggregate indicators should be called out and specified. If this data element is linked to an aggregate indicator, then indicate Yes. If not, indicate No.
Notes	If there is an issue or inconsistency in how a data field is defined, make a note of the issue. Irregularities and inconsistencies will need to be resolved at a later stage through a process of team discussion and triangulation. This column should also be used for any other notes, annotations or communication messages within the team.
Concept mappings	Depending on which systems you plan to interoperate with, other columns will most likely need to be added to map to the concepts used in the other system (e.g. ICD-11, SNOMED). One column should be used for each concept dictionary.

FHIR: Fast Health Interoperability Resources; ICD-11: International Classification of Diseases and Related Health Problems, 11th revision; SNOMED: Systematized Nomenclature of Medicine.

References

1. Integrated management of childhood illness: management of the sick young infant aged up to 2 months: IMCI chart booklet. Geneva: World Health Organization; 2019 (<https://iris.who.int/handle/10665/326448>).
2. *Classifying health workers: mapping occupations to the international standard classification*. Geneva: World Health Organization; 2019.



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